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Big Data Management & Analytics Master

Leveraging Vision Language Models for Multi-Modal Information Extraction and Knowledge Graph Construction

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Source code and implementation repository[1]

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Abstract

This thesis investigates the automated extraction and structuring of multimodal knowledge from unstructured French technical reports on hydro-ecological restoration in the Rhine basin, produced within the Franco-German TETRA project. These documents combine narrative text with maps, hydro-thermal graphs, and site photographs, yet remain difficult to search and reuse semantically.

The proposed system combines layout-aware document parsing, cross-modal image–text retrieval, grounded visual captioning, and ontology-based knowledge graph construction. Its design is evaluated through retrieval, captioning, and knowledge-enrichment experiments across alternative model architectures, prompting strategies, and context-integration configurations. The results show that document-specialised retrieval and structured, retrieval-grounded prompting increased the reported verified-enrichment proxy for most evaluated models, while changing the trade-off with the automated visual-consistency score. Parameter-efficient fine-tuning further improves the structural conformity of the generated knowledge graph to the target ontology. The resulting framework provides an implemented basis for transforming heterogeneous technical reports into structured, queryable multimodal knowledge for the TETRA digital toolbox. Its limitations include internally annotated ground truth, incomplete ontology coverage, and the absence of a completed manual audit of factual triple precision.

Keywords: multimodal information extraction; document layout analysis; cross-modal retrieval; vision-language models; grounded image captioning; knowledge graph construction; hallucination mitigation; parameter-efficient fine-tuning.

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Chapter 1

Introduction

The rapid evolution of deep learning architectures, particularly multi-modal Vision-Language Models, has introduced transformative capabilities for extracting, interpreting, and structuring knowledge from complex, heterogeneous document repositories. In highly specialized scientific and engineering domains, such as environmental engineering and hydro-ecological management, decades of field monitoring, geomorphological surveys, and river naturalization campaigns have generated vast catalogs of documentation. These catalogs containing technical dossiers, multi-temporal cartographic maps, engineering cross-sections, and annotated site photographs remain largely unstructured, isolated across regional institutions, and closed to automated semantic querying. Transforming this dense, multi-modal data into machine-interpretable, standardized knowledge representations is an urgent scientific milestone and a practical necessity for driving evidence-based decision-making in global environmental governance.

General Context

Sustainable river restoration and the ecosystem-led naturalization represents a critical environmental priority for international water policy. The Rhine River basin, particularly along the Franco-German border, serves as a premier example of this socio-technical challenge. Nearly two centuries of structural interventions that include mechanical channel rectification, industrial damming, embankment construction, and chemical pollution have drastically disrupted the river's natural hydromorphological dynamics, sediment transport loops, and native aquatic biomes.

To mitigate these disruptions, several cross-border ecological restoration campaigns have been executed and documented in technical reports, post-intervention audits, and longitudinal monitoring profiles. These environmental documents combine rich textual narratives with visual materials, such as geographic information system (GIS) maps, on-site photographs of restoration works, hydro-thermal graphs, and biological species distributions. However, this domain knowledge remains fundamentally isolated, heavily multi-lingual, and unorganized for comparative meta-analysis.

This research was conducted within the TETRA project, a Franco-German collaboration developing tools and methods to support the adoption of artificial intelligence in water management. TETRA evaluates these tools through use cases in river-water-quality monitoring and river restoration. Within this context, this thesis designs, implements, and evaluates an end-to-end pipeline for extracting knowledge from unstructured French hydro-ecological restoration reports and representing it in a schema-compliant knowledge graph.

Problem and Challenges

Automated information extraction from technical environmental documents presents several interconnected technical bottlenecks that render conventional document processing pipelines ineffective. Below are some of the bottlenecks that were addressed in this work:

1. **PDF Graphic Stream Decomposition:** Conventional data extractors parse document object trees sequentially. However, portable document formats (PDFs) store high-resolution GIS maps and vector drawings as collections of independent, multi-layered vector primitives and tiled raster sub-sheets. This structural fragmentation causes traditional parsers to segment a single coherent map into numerous separate, uninterpretable image shards, thereby destroying vital visual context.

2. **Single-Vector Dense Retrieval Bottlenecks:** Established cross-modal retrieval models, such as standard CLIP implementations[1], rely on a single-vector bi-encoder architecture that compresses an entire page or image asset into a single global embedding vector, discarding the fine-grained spatial and textual detail needed for precise document retrieval. While newer single-vector models built on large vision-language backbones partially mitigate this by encoding richer representations before compression, the architectural bottleneck remains most acute in generic contrastive encoders that were never optimized for dense, text-heavy visual content.

3. **Low-Resolution Scale-Collapse and Hallucination:** General-purpose VLMs frequently downscale input graphics into small, fixed pixel grids such as 224×224 pixels. This downsampling causes technical charts, scales, and legends to collapse into unreadable pixel noise, which forces the model's auto-regressive decoding loops to ignore the visual evidence and hallucinate data based entirely on linguistic pre-training priors [2].

4. **Cross-Modal Narrative Duplication:** Traditional textual extraction procedures frequently capture layout captions once inside the paragraph array and a second time within the image crop boundary. This repetition creates massive data redundancy that negatively impacts downstream indexing operations.

Objectives

Building upon prior work on textual knowledge representation at the ICube laboratory focused on automated case-base population from French river narratives [3], this project expands that framework into the multi-modal domain. The ultimate objective is to design a unified system where textual content and visual coordinates are resolved into a single, unified knowledge space. Concretely, this work pursues four primary technical objectives:

- **Layout-Aware Document Parsing:** Construct a computer-vision-driven document layout segmentation engine capable of isolating unified visual assets while eliminating vector fragmentation and redundant caption capture [4].

- **Cross-Modal Contextual Retrieval:** Investigate and evaluate retrieval strategies for associating independent visual charts with their corresponding textual descriptions in the document body (2), including both document-specialised dense retrieval and multi-vector late-interaction approaches [5, 6].
- **Two-Phase Hallucination Guardrailing:** Design a prompting architecture that separates visual inventory compilation from contextual interpretation to suppress text-bias hallucinations[7] over dense scientific graphics (3).
- **Schema-Compliant Graph Construction:** Programmatically map extracted textual and visual facts into standardized Resource Description Framework (RDF) triples that consistently populate an existing OWL hydro-geomorphological ontology [8].

Main Contributions

This thesis delivers the following verified engineering and architectural contributions to the field of multi-modal document intelligence:

- **Layout-aware image extraction:** Engineered a layout-aware processing loop combining high-fidelity page rasterization with a layout-aware extraction pipeline built on the pretrained `DocLayout-YOLOv10` object detector, featuring a custom geometric box-merging algorithm and a multi-tier text filtration stack to suppress institutional boilerplate noise.
- **Cross-modal contextual retrieval:** Evaluated document-specialised dense retrieval and multi-vector late-interaction retrieval strategies for associating independent visual charts with relevant textual descriptions in the document body (2). The evaluation identified the strongest retrieval configuration under the reported retrieval metrics, providing the textual context used by the downstream captioning pipeline [5, 6].
- **Image captioning ablation:** Conducted systematic ablation studies over different multi-modal models under three distinct context-fusion topologies, establishing a pixel-first prompting standard grounded in visual mitigation theory to substantially reduce language confirmation bias.
- **Multi-modal knowledge graph construction:** Formulated a programmatic serialization pipeline using `rdflib` that decouples generative triple extraction from file compilation, resolving cross-modal entities through deterministic label normalization and a three-stage entity-resolution process (exact-label merging, LLM-mediated alias resolution, and similarity-based alias merging) to fuse text-derived and image-derived facts into an integrated knowledge graph.

Research Objectives and Questions

The experimental evaluation is structured around four research questions that correspond to the four processing pipelines and define the criteria used to assess the framework on

the evaluation corpus:

1. **RQ1 : Layout-aware extraction:** To what extent does layout-aware rasterization and structural filtering produce coherent visual assets and reduce non-informative document content in French hydro-environmental reports? (refer Section 5.5)
2. **RQ2 : Cross-modal retrieval:** Which evaluated cross-modal retrieval configuration most reliably retrieves the narrative text blocks associated with extracted figures? (refer Section 5.6)
3. **RQ3 : Grounded image captioning:** How do structured prompting, retrieval-grounded context, context-fusion strategies, prompt-engineering strategies, and contrastive decoding affect the trade-off between verified enrichment, VQA-based visual consistency, and observed end-to-end throughput? (refer Section 5.7.1 through Section 5.7.7)
4. **RQ4 : Multi-modal knowledge-graph enrichment:** To what extent do image-derived captions contribute to the enrichment and cross-modal integration of a text-derived knowledge graph, and how does predicate-focused fine-tuning affect ontology-schema conformance? (refer Section 5.8)

Outline

The remainder of this report is organized as follows:

- **Chapter 2: Related Work** conducts a comprehensive review of the state-of-the-art literature in document layout analysis, cross-modal document retrieval, vision-language prompting strategies, and multi-modal knowledge graph population frameworks.
- **Chapter 3: Technical Background** introduces the foundational concepts underlying document analysis, cross-modal retrieval, vision-language generation, and knowledge-graph construction.
- **Chapter 4: Proposed Methodology** details the complete architectural formulation, operational parameters, and algorithmic pseudocode across all four core pipelines.
- **Chapter 5: Experimental Evaluation and Ablation Analysis** presents the evaluation metrics, experimental protocols, and results for the four pipelines, including layout extraction, image–text retrieval, grounded image captioning, and ontology-based knowledge graph construction. It reports the corresponding ablation studies and the multi-criteria selection of the final pipeline configuration.
- **Chapter 6: Conclusion and Future Work** summarizes the primary findings, reviews active limitations, and maps out concrete development vectors for future expansions within the TETRA project toolbox.

Chapter 2

Related Work

This chapter presents a systematic, comparative review of the state-of-the-art literature, foundational methodologies, and historical research trajectories relevant to multimodal document intelligence and semantic knowledge extraction. In accordance with the project objectives, the reviewed literature is organized into four structural categories: document layout analysis and visual parsing (2.1), cross-modal document retrieval and matching (2.2), prompting and hallucination mitigation (2.3), and ontology population and knowledge graph construction (2.4). For each category, the underlying core concepts are analysed along with identifying systematic limitations and explicitly justifying the architectural choices.

2.1 Document Layout Analysis and Visual Parsing

2.1.1 Main Ideas and Core Paradigms

Document Layout Analysis (DLA) is the critical first stage for structuring raw, multimodal digital assets. Early approaches relied heavily on programmatic text stream parsers (e.g., native `PyMuPDF` or `PDFMiner` coordinate groupers). These tools decode the internal page object trees sequentially, grouping text characters into paragraphs based on spatial proximity heuristics.

To overcome the limitations of these text-blind methods, information extraction has pivoted towards vision-driven document layout segmentation. Early deep learning models framed layout analysis as a semantic segmentation or object detection task, utilizing deep Convolutional Neural Networks (CNNs) such as Mask R-CNN[9] or Faster R-CNN[10] trained on large-scale open-source datasets like PubLayNet[11] and DocLayNet[4]. Recent document-layout systems use object-detection architectures adapted to document structures. One example is `DocLayout-YOLO`, which adapts the YOLOv10 detection architecture to document layout analysis [12, 13]. By eliminating non-maximum suppression (NMS) bottlenecks through consistent dual-label assignments during training and processing pages as flattened pixel matrices at high resolutions (1024×1024 pixels), these networks predict semantic bounding boxes based on global visual layouts rather than internal file code streams.

2.1.2 Limitations in the Target Domain

Traditional text-stream parsers exhibit a fundamental flaw when encountering complex technical or environmental engineering reports. Within these documents, high-resolution GIS maps, river morphology cross-sections, and charts are stored internally within the PDF format as deeply layered, separate vector paths, floating annotations, and independent

`XObject` raster sub-sheets. Lacking visual awareness, a stream parser can split a visually unified figure into fragmented image shards, disconnecting map grids from their legends and scales. Conversely, while early CNN-based layout models preserved visual coherence, they carried heavy computational overhead during mask rendering and frequently suffered from blurred bounding box boundaries around compact elements like figure captions.

2.1.3 Justification of Direction

The proposed approach uses full-page visual rasterization with `DocLayout-YOLOv10` rather than stream-level parsing, preserving vector layers and text labels as a single visual composition. Geometric block merging and caption absorption are then used to reduce edge truncation and retain figure-associated context.

2.2 Cross-Modal Document Retrieval and Matching

2.2.1 Main Ideas and Core Paradigms

Once textual and visual assets are isolated from document streams, cross-modal indexing is required to pair images with their corresponding narrative descriptions. The literature splits this task into two primary architectural families, which are Single-Vector Dense Bi-Encoders and Multi-Vector Late-Interaction Retrieval Networks.

2.2.1.1 Family 1: Single-Vector Dense Retrieval

Single-vector dense retrievers encode an image and a text block independently into a shared embedding space and estimate their semantic correspondence through similarity scoring. CLIP[1] is a widely used example, while more recent document-oriented models such as DSE-Qwen2[14] and VDR[5] use vision-language backbones to represent visually complex document pages. Their main advantage is efficient retrieval over large candidate sets; their limitation is that compressing an entire image or text block into one vector can lose localized layout and semantic detail.

2.2.1.2 Family 2: Multi-Vector Late-Interaction Retrieval

Multi-vector late-interaction retrievers preserve multiple token- or patch-level representations and compare them at query time, rather than reducing each modality to a single global vector. ColBERT-style late interaction [15] has been extended to document retrieval through models such as `ColPali` [6], `ColQwen2` [16], and `Nemotron ColEmbed` [17]. This family can retain local text, visual, and layout information more effectively than single-vector methods, but generally introduces higher computational and storage costs.

2.2.2 Limitations in the Target Domain

Standard CLIP-style single-vector bi-encoders (Family 1) suffer from an information bottleneck. Compressing an entire complex page layout, historical map, or multi-panel data chart into a single global vector flattens localized text features and layout configurations. This compression results in low-confidence alignments and narrow scoring margins when

distinguishing between highly specific technical passages. Conversely, while multi-vector late-interaction networks (Family 2) preserve exceptional visual detail, their standard scoring formulations sum over the entire token count of the text query, introducing a systematic mathematical bias that penalizes concise text blocks and misaligns text-length variations. Recent single-vector retrievers built on vision-language backbones partially mitigate this bottleneck by encoding richer layout and text detail before compression, but they still require empirical validation in this domain.

2.2.3 Justification of Direction

This thesis evaluates both single-vector and multi-vector retrieval families because their relative performance depends on the visual complexity of the target documents, the granularity of text blocks, and operational cost. For multi-vector models, mean-normalized MaxSim is used to reduce the influence of variable text-block length, while block-level matching is used instead of full-page aggregation to preserve localized context. OCR text is excluded from the retrieval query because preliminary tests showed that it encouraged literal string matching at the expense of visual-layout matching.

2.3 Prompting and Hallucination Mitigation

2.3.1 Main Ideas and Core Paradigms

Generating structured domain descriptions from technical figures requires robust VLMs. However, autoregressive multi-modal networks are susceptible to semantic hallucinations when parsing complex scientific graphics. The literature identifies **Linguistic Confirmation Bias** as a primary driver of these errors. When textual context is provided to a model prior to the evaluation of visual layers, the self-attention weights prioritize textual terms, which causes the vision component to generate features that match textual expectations rather than the actual pixel data.

To mitigate this bias, contemporary prompting research focuses on enforcing strict pixel grounding. The **Visual Description Grounded Decoding (VDGD)** [7] framework generates a detailed textual description of the image prior to response generation, then samples output tokens based on their divergence from that description, requiring the model to ground its answers in an explicit visual account before performing contextual inference. This approach is supported by the **Multi-Modal Mutual-Information Decoding (M3ID)** architecture [18], which amplifies the influence of visual evidence over language priors during decoding to prevent the model from conforming to pre-training language expectations during extended generation passes. Concurrently, **Black-Box Visual Prompt Engineering** frameworks [19] demonstrate that overlaying visual markers (e.g., bounding boxes, circles) directly on the image, selected via a learned router model, effectively stabilizes internal multi-modal attention layers without requiring access to model internals.

2.3.2 Limitations in the Target Domain

Standard zero-shot visual captioning models, such as BLIP[20] and BLIP-2[2], lack localized optical character recognition (OCR) capabilities and generate short, generic descriptions unsuitable for technical graphics. Furthermore, standard prompt strategies that inject raw text references or OCR tokens before the image parsing stage consistently activate language confirmation bias, causing the model to misinterpret text concepts over the visual fields.

2.3.3 Justification of Direction

The proposed captioning approach uses a high-resolution vision-language model with a two-phase pixel-first prompting strategy. Phase 1 restricts generation to image-based observation before contextual information is introduced. Phase 2 then uses retrieved local context and document-level information to identify supported additions while limiting language confirmation bias.

2.4 Ontology Population and KG Construction

2.4.1 Main Ideas and Core Paradigms

Ontology population converts unstructured content into schema-compliant knowledge graph assertions. Earlier approaches based on dependency parsing or static transformer models map textual content to RDF triples, while generative language models can extract relational facts more flexibly but may produce malformed RDF syntax.

Recent work has explored the use of large language models for ontology population in specialized domains [21]. In parallel, multimodal knowledge graph research investigates how textual and visual evidence can be integrated to enrich entity and relation representations [22]. Frameworks such as VaLiK [23] further examine the construction of multimodal knowledge graphs from visual and textual inputs without manual text annotations. These approaches motivate the integration of image-derived evidence into ontology-guided knowledge extraction, while leaving the document-specific challenges of layout parsing, localized retrieval, and schema-constrained RDF serialization to be addressed.

Prior work by Ghazouani, Giustozzi, and Le Ber [3, 24] demonstrated ontology-guided extraction from unstructured text using schema-constrained prompting and fine-tuning on synthetic data. Their work showed that ontology information can guide relational triple extraction, but it was limited to textual evidence and did not address figures, maps, or other document-layout modalities.

2.4.2 Limitations of Prior Works

While the methodologies established by Ghazouani et al. [3, 24] achieve high precision when extracting structured triplets from text, they do not account for the visual and layout modalities of document structures. Consequently, embedded engineering diagrams, multi-temporal cartographic maps, and site photographs are treated as context gaps, leaving vital visual data completely unindexed.

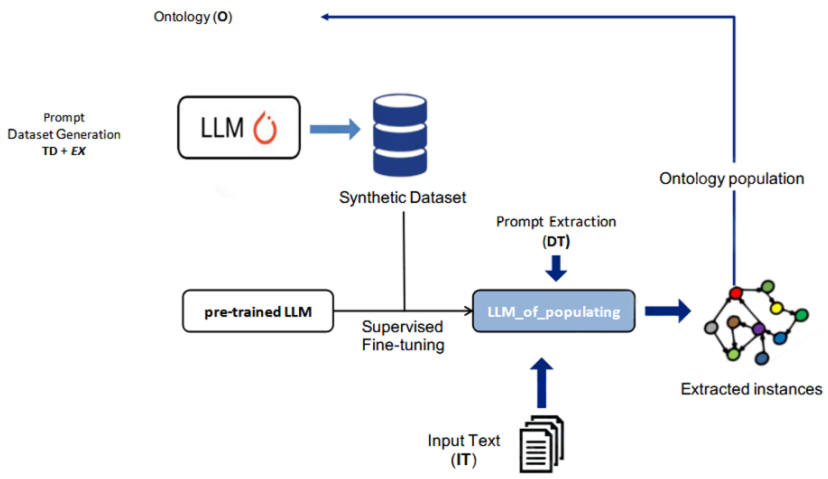


Figure 2.1: Fine-tuning pipeline for ontology-guided triplet extraction, comprising (1) prompt-based synthetic dataset generation, (2) supervised fine-tuning of a pre-trained LLM, and (3) inference-time ontology population using the fine-tuned model. Adapted from Ghazouani et al. [24]

2.4.3 Justification of Direction

The fine-tuned text-only ontology population approach (Fig: 2.1) of Ghazouani et al. [3, 24] demonstrate the value of schema-constrained prompting and fine-tuning for relational extraction. This thesis extends that direction by incorporating visual evidence and separating generative fact extraction from deterministic RDF serialization to reduce malformed graph output.

2.5 Summary and Technical Background

The reviewed literature shows substantial progress in document layout analysis, cross-modal retrieval, hallucination mitigation, and text-based ontology population. However, these advances are generally treated as separate tasks, and prior ontology population approaches do not integrate visual and layout evidence with text-derived facts in a unified graph-construction process. Chapter 3 introduces the technical background required for the proposed multi-modal framework.

Chapter 3

Background

This chapter introduces the foundational concepts, network architectures, processing algorithms, and mathematical notations required to operationalize the methodology presented in Chapter 4. Each concept is paired with a formal definition, structured notation, and a concrete domain example to ground the theoretical framework within technical document analysis.

3.1 Notation

Throughout this report, a consistent mathematical notation is maintained across all chapters. Scalars are represented by standard lowercase letters such as s, vectors are denoted by bold lowercase letters such as $\mathbf{v} \in \mathbb{R}^d$, and matrices are signified by bold uppercase letters such as $\mathbf{W} \in \mathbb{R}^{d \times d'}$. An individual document page is denoted by p, a textual query string by q, and an isolated image asset by $\mathcal{I}_m$. The complete set of document pages is defined as $\mathcal{P} = \{p_1, p_2, \ldots, p_N\}$, whereas the corresponding set of extracted visual images is defined as $\mathbb{I} = \{\mathcal{I}_1, \ldots, \mathcal{I}_M\}$. A formal knowledge graph structure is expressed as $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$, where $\mathcal{E}$ represents the bounded set of entity nodes, $\mathcal{R}$ denotes the set of relational predicates, and $\mathcal{T} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$ constitutes the complete set of populated semantic triples.

3.2 Transformer Architecture and Self-Attention

The vision-language models evaluated within this framework are constructed upon the foundational Transformer architecture [25]. Consequently, mapping the mathematical mechanics of the self-attention layer is essential to understanding multi-modal feature alignment.

3.2.1 Self-Attention

Self-attention derives a contextualized representation for each token in a sequence by computing an attention distribution over the entire sequence. Each input embedding is projected into query, key, and value vectors via learned weight matrices; the query and key vectors determine how strongly a token attends to every other token, via a scaled dot-product softmax, while the value vectors are combined according to these attention weights to produce the token's contextualized output. This mechanism underlies the vision-language backbones evaluated in this report (Sections 4.3 and 4.4), where it governs how visual and textual tokens are related during cross-modal encoding.

3.2.2 Vision Transformer (ViT)

To process visual images using sequence-to-sequence architectures, the Vision Transformer model partitions an input image asset defined by $\mathcal{I} \in \mathbb{R}^{H \times W \times C}$ into a series of non-overlapping image patches of fixed size $P \times P$, which yields a total of $N_p = HW/P^2$ patches [26]. Each individual patch is flattened into a one-dimensional array and linearly projected into a d-dimensional embedding space. A learnable positional embedding tensor is subsequently added to the patch sequence to preserve original two-dimensional spatial structures:

$$\mathbf{z}_0 = [\mathbf{e}_{cls};\ \mathbf{E}\mathbf{p}_1;\ \mathbf{E}\mathbf{p}_2;\ \ldots;\ \mathbf{E}\mathbf{p}_{N_p}] + \mathbf{E}_{pos},$$

where $\mathbf{p}_i$ represents the i-th flattened image patch, $\mathbf{E}$ denotes the trained patch projection matrix, $\mathbf{e}_{cls}$ signifies a learnable classification token, and $\mathbf{E}_{pos}$ stores the absolute spatial coordinates. The output state corresponding to the classification token represents the global image embedding vector.

Example: A target document image measuring 224×224 pixels processed with a patch size configuration of $P = 16$ results in a sequence of $N_p = (224/16)^2 = 196$ patch tokens, where each token encodes the visual properties of a localized 16×16 pixel bounding region.

3.3 CLIP: Contrastive Language-Image Pretraining

The Contrastive Language-Image Pretraining (CLIP)[1] framework jointly optimizes an image encoder f_I and a textual encoder f_T over a large corpus of image-text pairings using a symmetric contrastive objective function. Given a training batch containing N valid pairs, the objective function maximizes the cosine similarity scores of correct cross-modal matches while simultaneously minimizing the alignment scores of the remaining $N^2 - N$ incorrect combinations:

$$\mathbf{S}_{ij} = \frac{f_I(\mathcal{I}_i) \cdot f_T(t_j)}{\|f_I(\mathcal{I}_i)\| \cdot \|f_T(t_j)\|}.$$

During inference, a target textual query q and a candidate set of document images $\{\mathcal{I}_k\}_{k=1}^{K}$ are encoded independently, and the highest scoring cross-modal match is selected using the optimization parameter $\arg\max_k \cos(f_T(q), f_I(\mathcal{I}_k))$.

Example: Given a document figure showing a riverbank and a text string reading "Figure 14: Erosion de berge à l'amont du Vieux Rhin", the model calculates a single scalar cosine similarity score between the global image vector and the global text vector to evaluate their semantic correspondence (see Figure 4.5).

3.4 Multi-Vector Retrieval and Late Interaction

3.4.1 Limitation of Single-Vector Retrieval

Single-vector retrieval models compress an entire page layout or visual image into a single global embedding vector. Although computationally efficient, this structural compression collapses local spatial configurations and fine-grained semantic features into a single point, which restricts the architecture's capacity to align specific visual regions with individual text tokens [6].

3.4.2 ColBERT-Style Late Interaction

Multi-vector late-interaction architectures preserve individual per-patch embeddings and defer the cross-token comparative alignment step until query execution time [15]. Given a set of query token embedding vectors defined as $\mathbf{Q} = \{\mathbf{q}_i\}_{i=1}^{|Q|}$ and a corresponding set of document image patch embeddings defined as $\mathbf{D} = \{\mathbf{d}_j\}_{j=1}^{|D|}$, the maximum similarity score is computed as follows:

$$\text{Score}(\mathbf{Q}, \mathbf{D}) = \sum_{i=1}^{|Q|} \max_{j \in |D|} \cos(\mathbf{q}_i, \mathbf{d}_j).$$

For each individual query token, the system identifies the most semantically similar document patch, and the summation of these maximum local matches determines the global relevance score [6]. This token-level matching allows a specific query token encoding the term "fontaine" to align with the precise image patch containing that visual text element, regardless of the contextual composition of the surrounding image patches (see Figure 4.5).

3.4.3 Mean-Normalized MaxSim

The standard MaxSim scoring formulation scales naturally with query length because it sums across the total number of query tokens. To account for variable-length text passages, a mean-normalized variation is implemented to stabilize the relevance evaluation across diverse sentence structures:

$$\text{Score}_{\text{norm}}(\mathbf{Q}, \mathbf{D}) = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \max_{j \in |D|} \cos(\mathbf{q}_i, \mathbf{d}_j).$$

Example: The ColPali model segments an input document page layout into a visual grid that yields approximately 1024 distinct patch embeddings [6]. A textual query reading "érosion de berge" is processed into roughly 4 tokens. The late-interaction mechanism isolates the document patch that maximizes the cosine similarity for each individual query token, and the average sum of these four local scores determines the final relevance ranking.

3.5 Vision-Language Architecture Evolution

To contextualize the multi-modal extraction capabilities of modern conversational models, it is necessary to examine the evolution of visual token processing paradigms, contrasting structural query compression with adaptive native resolution architectures.

3.5.1 Token Compression and Q-Former

Early modular vision-language architectures, demonstrated by the BLIP-2[2] framework, implement an intermediate alignment layer to connect static visual encoders with static language models. This paradigm uses a lightweight Querying Transformer, which acts as an information mediator that extracts language-relevant features from visual representations and projects them into the input token space of the language network.

The alignment module maintains a fixed configuration of $N_q = 32$ learnable query tokens defined as $\{\mathbf{q}_1, \ldots, \mathbf{q}_{N_q}\}$. These queries interact with the frozen image encoder's patch features using cross-attention mechanisms, selectively isolating the details that align with linguistic concepts:

$$\hat{\mathbf{q}}_i = \text{CrossAttn}(\mathbf{q}_i,\ \mathbf{V}_{enc}),$$

where $\mathbf{V}_{enc}$ represents the output feature tensor generated by the static visual encoder. The 32 distilled visual representations are subsequently projected through a linear layer into the input dimensionality of the language model to guide autoregressive text generation.

Example: Given an image depicting an engineering dam structure, the alignment tokens isolate specific semantic features such as "water level" and "concrete barrier" while bypassing uninformative background context, compressing the entire asset into 32 distilled tokens before sequence generation.

3.5.2 Native and Dynamic Resolution Encoding

While fixed-token or uniform grid-resizing configurations downscale input document pages to standard dimensions like 224×224 pixels, this spatial compression can obscure fine-grained configurations, small text fonts, and annotations within engineering maps or charts.

To mitigate this layout breakdown, alternative architectures process image assets at their native resolution and original aspect ratio. The Pixtral-12B architecture implements two-dimensional rotary position embeddings to process images without destructive resizing operations, generating a variable number of patch tokens that scale proportionally with the original image dimensions [27]. Similarly, the Qwen2-VL model utilizes a dynamic resolution mechanism that preserves variable aspect ratios by partitioning document screenshots into a flexible number of visual tokens determined by the structural complexity of the input source [16].

3.6 Knowledge Representation

3.6.1 Ontologies and Knowledge Graphs

An ontology is conventionally understood as a formal, explicit specification of a shared conceptualization of a given domain [24]. It distinguishes between the Terminological Box (TBox), which encodes the schema of the domain (class hierarchy, relation definitions, and axioms), and the Assertional Box (ABox), which encodes concrete facts about specific individuals. The TBox is established during initial modeling, while the ABox is enriched during ontology population, allowing the ontology to represent real-world data and support downstream reasoning.

Once populated, an ontology is generally referred to as a knowledge graph, formally a tuple $G = (E, R, F)$, where E is the set of entities, R is the set of relation types, and $F \subseteq E \times R \times E$ is the set of factual assertions [24]. Each fact is a triplet (s, r, o), with $s, o \in E$ the subject and object entities and $r \in R$ the relation. This formulation clarifies the relationship between ontology population and Relation Triplet Extraction

(RTE): given an input textual unit S_i, RTE jointly identifies a triplet (s_i, r_i, o_i) drawn from the target ontology's relation set R. Framing ontology population as an RTE task is central to the methodology developed in this thesis, since it allows both textual and visual evidence to be reduced to a common target representation before being merged into a unified knowledge graph, as detailed in Chapter 4.

3.6.2 RDF, OWL, SKOS, and SHACL

The Resource Description Framework (RDF) materializes the triplet formalism (s, r, o) as a directed, labeled graph in which subjects and predicates are identified by Internationalized Resource Identifiers (IRIs), while objects may be either IRIs or literal values [28]. The Web Ontology Language (OWL) extends RDF with expressive semantic constraints, operationalizing the TBox through constructs such as `owl:Class`, `owl:NamedIndividual`, `owl:ObjectProperty`, and `owl:DatatypeProperty` [29]. The Simple Knowledge Organization System (SKOS) provides a standardized vocabulary for concept hierarchies and taxonomies via properties such as `skos:broader` and `skos:narrower` [30], while the Shapes Constraint Language (SHACL) validates populated RDF graphs against predefined structural shapes to ensure schema compliance [31]. SPARQL is the standard query language for extracting information from RDF triple stores [32]. **Illustrative RDF example.** The

following Turtle fragment illustrates how information from a river-restoration document can be represented in RDF. It identifies a radier as an individual, assigns a human-readable French label, and links it to its documented purpose:

```
@prefix : <http://www.semanticweb.org/fethi/ontologies/
2024/1/TetraOnto.owl#> .
@prefix owl: <http://www.w3.org/2002/07/owl#> .
@prefix skos: <http://www.w3.org/2004/02/skos/core#> .

:Radiers a owl:NamedIndividual ;
    skos:prefLabel "radiers"@fr ;
    :hasPurpose "prévenir le risque d'érosion régressive"@fr .
```

Here, `:Radiers` is the subject entity, `:hasPurpose` is an ontology-defined predicate, and the French text literal is the object. Turtle prefixes provide a compact notation for full IRIs, while semicolons group multiple assertions about the same subject.

Section 4.5.3 details how these technologies are concretely instantiated through the TetraOnto domain schema used throughout this thesis. This chapter has detailed the theoretical foundations and algorithms driving the individual components of our multi-modal data processing framework. Chapter 4 builds upon these concepts, and in particular on the RTE formulation of ontology population introduced above, to detail how these models are integrated to address the specific challenges of multi-modal information extraction from unstructured French environmental documents.

Chapter 4

Our Methodology and Approach

This chapter presents the four-pipeline system developed to extract, match, caption, serialize, and fuse multi-modal knowledge from French technical hydro-environmental reports into an ontology-conditioned knowledge graph. The architecture is designed to address non-linear page layouts, embedded geographic information system (GIS) maps, technical diagrams, and structural relational sparsity natively. Section 4.1 outlines the global system workflow, while Sections 4.2 through 4.5 detail the algorithmic components, feature-transformation layers, and mathematical formulations governing each module. The source code and implementation guidelines required to reproduce the experimental configurations are available in the project repository.[1]

4.1 System Overview

The multi-modal knowledge extraction framework is organized into four sequential and dependent processing pipelines:

- **Pipeline 1: Document Layout Parsing and Image Extraction**: Processes unstructured technical documents, segments page layouts using visual parsing models, and extracts text blocks and figure regions with their page coordinates.
- **Pipeline 2: Semantic Contextual Retrieval**: Ranks text blocks for each extracted image using cross-modal similarity scores to identify the most relevant narrative context.
- **Pipeline 3: Grounded Visual Captioning**: Uses vision-language models and a two-phase prompt to generate domain-specific descriptions of technical figures, with visual verification and retrieved context.
- **Pipeline 4: Cross-Modal Semantic Integration**: Converts textual narratives and verified visual captions into schema-compliant Resource Description Framework (RDF) triples, then resolves cross-modal entity duplicates through label normalization and a three-stage entity-resolution process.

Each independent pipeline can be executed and validated modularly, allowing for structured component optimization. Figure 4.1 illustrates the unified multi-modal knowledge graph population architecture.

[1] Project source code on GitHub

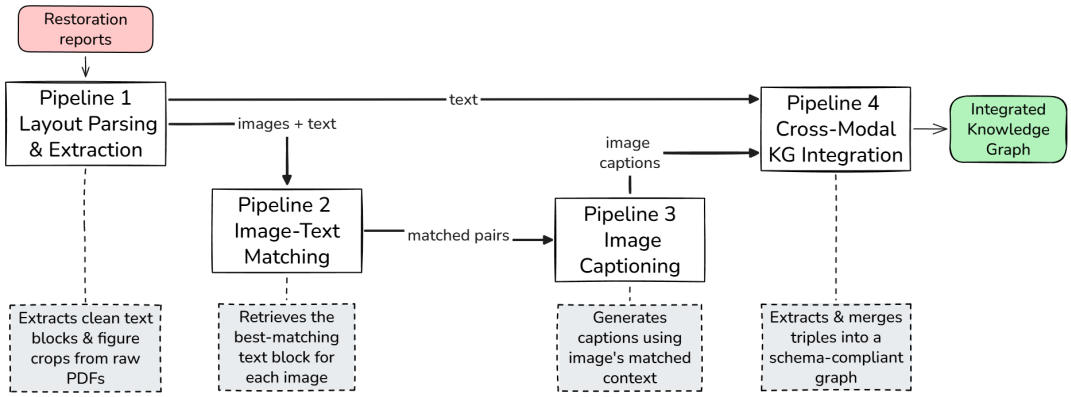


Figure 4.1: Overall four-pipeline system architecture for multi-modal knowledge graph construction from French Rhine restoration reports

4.2 Pipeline 1: Layout Parsing and Cross-Modal Extraction

The foundational module of the multi-modal knowledge graph population architecture is a layout-aware document extraction and semantic cleaning pipeline. Environmental and technical engineering dossiers exhibit a high density of non-linear layouts, multi-temporal cartographic series, complex vector diagrams, and tabular schemas. Traditional stream-based text parsers process these documents sequentially, failing to associate embedded figures with their textual references and polluting the output with non-informative institutional noise such as running headers, watermarks, and corporate logos. Pipeline 1 uses a multi-stage extraction process to isolate text blocks and visual regions while retaining their page-level spatial relationships, as illustrated in Figure 4.2. Each of the components depicted in this architecture is described in detail in the following subsections.

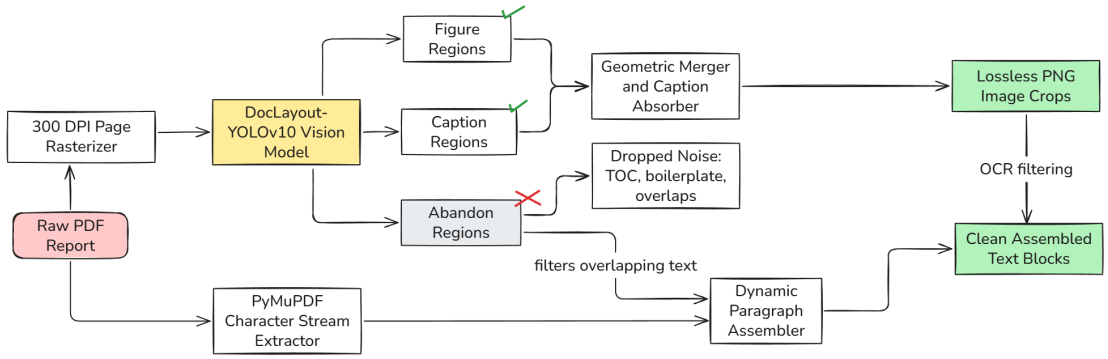


Figure 4.2: Architectural overview of Pipeline 1, illustrating page rasterization, DocLayout-YOLO semantic region detection, dynamic block assembly, and cross-page n-gram boilerplate filtration.

4.2.1 Limitations of Non-Visual Structural Parsing

Initial structural prototypes constructed during the exploratory phase of this work relied on traditional object-stream extraction models, utilizing native PDF dictionary parsing

and bounding-box grouping via the `PyMuPDF` library [33]. This approach parsed the underlying page object trees directly, extracting embedded graphical elements based on internal `XObject` reference keys.

Initial tests on the environmental-document corpus showed that this approach fragmented complex figures. PDFs render high-resolution GIS maps, river morphology cross-sections, and complex historical charts as multi-layered collections of independent vector primitives, tiled raster sub-sheets, and detached text labels. Lacking structural and visual awareness, the object-stream extraction pass breaks individual figures into fragmented tiles that sever map bodies from their accompanying legends and scales. As illustrated in Figure 4.3(a–b), the legacy object-stream extractor separated a representative cartographic figure into two disconnected fragments. The corrected layout-aware pipeline subsequently recovered the same source figure as one coherent visual asset, preserving the map body, legend, scale, and caption, as shown in Figure 4.3(c). This failure mode was observed repeatedly across the corpus, including a more severe case in which a single multi-panel photographic figure was broken into sixteen separate fragments. This structural fragmentation destroys the semantic and visual context required for downstream multi-modal retrieval or captioning. This limitation motivated the page-rasterization and layout-detection approach introduced in the following subsection.

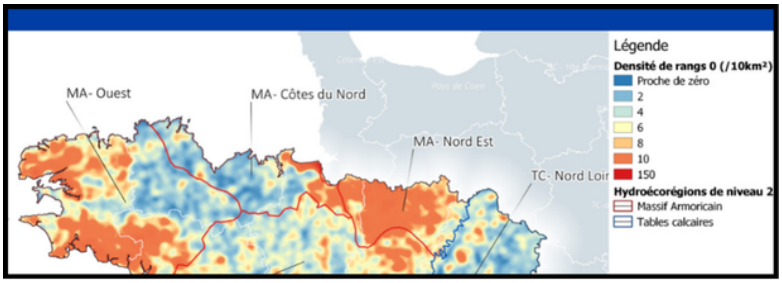
4.2.2 Visual Layout Segmentation and Geometric Consolidation

To preserve the visual context of scientific figures, the extraction process treats each PDF page as a two-dimensional image. The process begins by rasterizing pages as lossless PNG images at 300 DPI, flattening vector primitives and layered PDF objects into a single page image.

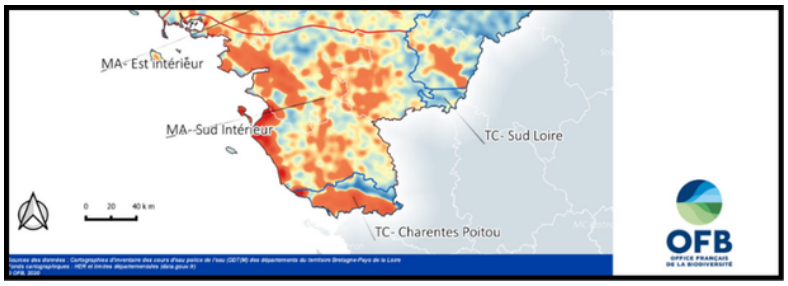
The flattened page images are processed in batches of size $B_{\text{yolo}} = 8$ through a fine-tuned document layout analysis network, `DocLayout-YOLOv10`, utilizing the specialized checkpoint trained on the DocStructBench benchmark [13]. Operating at an input dimension of 1024×1024 pixels and an empirical detection confidence threshold of $c_{\min} = 0.20$, the model maps two-dimensional bounding box coordinates $b = [x_1, y_1, x_2, y_2]$ to specific document semantic classes, distinguishing between text paragraphs, figures, captions, and structural margins. Detections mapped to the target categories $\mathcal{C}_{\text{fig}} \in \{\texttt{figure}, \texttt{picture}, \texttt{image}\}$ are retained as figure regions, while elements mapped to $\mathcal{C}_{\text{cap}} \in \{\texttt{figure_caption}, \texttt{caption}\}$ are processed independently as layout anchors, corresponding respectively to the Figure Regions and Caption Regions branches in Figure 4.2. Bounding boxes categorized as `abandon` are recorded as regions to suppress, preventing non-informative visual zones such as page borders or decorative dividers from polluting the text corpus.

Because sub-components within a complex multi-panel figure can trigger multiple distinct bounding boxes, a geometric block-merging pass is applied to reconstruct complete visual compositions. Let W_{page} and H_{page} define the absolute width and height of the page plane. Two discrete figure bounding boxes b_i and b_j are merged into a single encompassing block if their minimum horizontal distance Δx and vertical distance Δy satisfy the following layout constraint:

$$\Delta x < 0.08 \times W_{\text{page}} \quad \text{and} \quad \Delta y < 0.02 \times H_{\text{page}}.$$



(a) First fragment produced by the legacy object-stream extractor.



(b) Second fragment produced from the same source figure.

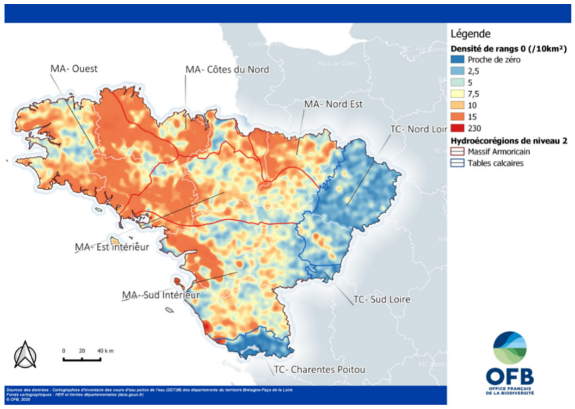


Figure 7 : Densité de rangs zéro maille de 10 km² sur le territoire Bretagne-Pays de la Loire d'après les cartographies départementales des cours d'eau au titre de la police de l'eau (Galineau, 2020)

(c) Unified cartographic figure produced by the corrected layout-aware extraction pipeline, preserving the map body, legend, scale, and source caption within one coherent visual asset.

Figure 4.3: Comparison between legacy object-stream extraction and the corrected layout-aware extraction approach for a cartographic figure from PDF-2 (Section 5.2). The legacy extractor splits the figure into disconnected fragments, separating visual content from contextual elements (a–b). The corrected Pipeline 1 output reconstructs the figure as a single coherent visual asset (c).

Once the primary figure boundaries are stabilized, an associative caption routine is executed. If a detected caption box b_{cap} exhibits positive horizontal overlap with a consolidated figure box b_{fig} and sits directly beneath its lower border within a vertical margin of $0.08 \times H_{\text{page}}$, the caption boundary is absorbed into the figure's spatial boundaries. Finally, each consolidated bounding box is expanded by 30 pixels on every side before cropping, reducing the risk that chart legends or axis labels are excluded. Representa-

tive examples of the final visual crops produced by the extraction pipeline are shown in Figure 4.4, including a photographic drainage-channel crop and a GIS map crop.



Figure 14 : Localisation des sites de transferts étudiés (en rouge) par rapport au site d'origine (en violet)

(a) Extracted GIS map crop showing transfer site locations relative to the origin site.



Figure 1-9 : Aspect du cours d'eau au droit des premières habitations sur le tronçon 4

(b) Extracted pair showing a culvert structure and vegetated drainage channel.

Figure 4.4: Representative examples of final image crops produced by Pipeline 1, illustrating the diversity of visual content correctly isolated from source PDFs: GIS map (left) and photographs (right).

4.2.3 Dynamic Block Assembly and Content Extraction

Simultaneously, corresponding to the PyMuPDF Character Stream Extractor and Dynamic Paragraph Assembler branches in Figure 4.2, a layout-aware programmatic text parser extracts underlying character sequences from the unflattened document stream, mapping each text token to its corresponding typeface attributes and layout coordinates. To suppress running page numbers and institutional footers at the layout level, a 2.5% vertical margin filter is enforced at the top and bottom of each page coordinate plane.

Characters are grouped into lines, and individual lines are dynamically assembled into logical paragraphs using five joint constraints. Let $\bar{h}$ represent the median line height computed globally for a given page layout, and let s_{body} and f_{body} represent the dominant font size and typeface family covering the greatest character area. The system initializes a dynamic vertical gap threshold defined as:

$$\tau_g = 1.4 \times \bar{h}.$$

A running text block is closed, and a new block is initialized, if any of the following conditions are met:

1. **Vertical Discontinuity:** The vertical gap between the lower boundary of line l_i and the upper boundary of line l_{i+1} exceeds the dynamic threshold τ_g.

2. **Margin Alignment Shift:** The left-margin coordinate x_1 of line l_{i+1} deviates from the left-margin coordinate of the current block by more than 5 pixels, indicating a structural paragraph indent or a column break.

3. **Typographic Inconsistency:** The line font size or typeface shifts away from the current block's baseline parameters, indicating a header or block quote. Any line whose font size satisfies $s_l > 1.25 \times s_{\text{body}}$ is flagged as structural text and prefixed with a `[HEADER]` token.

4. **Grammatical Completion:** A block boundary is forced if the terminal character of line l_i passes a sentence end validation check where the parameter is_sentence_ender evaluates to true. This verification step uses regular expressions to intercept French abbreviations such as *art., al., ch., sect., fig., tab., p. ex., et al.*, unmatched opening guillemets, or running prepositions and conjunctions, preventing paragraph splitting across line wraps. Hyphenated line breaks are resolved by stripping the trailing hyphen and joining the text segments seamlessly.

5. **Caption Boundary Triggering:** Any line matching primary French hydrological report caption prefixes automatically triggers a new block boundary to isolate captions from surrounding narrative paragraphs.

To maintain grammatical coherence across page breaks, a cross-page stitching module inspects the terminal text block of page p. If is_sentence_ender evaluates to false, the pipeline runs a look-ahead pass onto page $p + 1$, extracts the first non-empty text paragraph, splits it at its first terminal punctuation mark, and appends the leading segment directly to the end of page p. The consumed text on page $p + 1$ is replaced with an empty string (ε) to maintain index alignment with the parallel bounding-box lookup arrays.

4.2.4 Multi-Stage Noise Filtering and Cross-Modal Deduplication

Prior to saving the extracted data to disk, the text blocks undergo a multi-layer filtration sequence to eliminate structural noise, institutional boilerplate, and cross-modal redundancy, as summarized in Figure 4.2.

4.2.4.1 1. Cross-Page Noise Profile Extraction

Before writing blocks to the output schema, two global analysis sweeps scan the assembled text arrays of the entire document to isolate cross-page repetitive noise:

- **Block-Level Repeater Isolation:** Individual text blocks are normalized by converting characters to lowercase, stripping punctuation, and removing page indexes. Normalized strings that appear on at least 60% of total document pages are flagged as exact structural repeaters. A secondary fuzzy pass groups blocks by their initial 20 characters and applies a Gestalt pattern matching algorithm [34]. If the fuzzy similarity ratio meets or exceeds $T_{\text{fuzzy}} = 0.88$ across the same 60% page threshold, the blocks are flagged as near-identical repeaters, capturing running headers that vary only by section numbers.

- **Token-Level N-Gram Sweeping:** A two-pass sliding window sweeps the global document text stream. Long, fused footer strings and institutional identifiers are captured by collecting 6-grams present on at least 35% of total pages, followed by a secondary sweep collecting 3-grams present on at least 60% of total pages to catch short corporate headers. Words matching these flagged n-gram profiles are stripped out from all blocks.

4.2.4.2 2. Spatial and Structural Layout Pruning

During final block assembly, each paragraph is checked against five serial filters: first, empty text slots (ε) are skipped; second, blocks under 160 characters that contain dot-leaders followed by numeric integers are discarded as Table of Contents entries; third, blocks matching the global repeater set are dropped; fourth, blocks whose spatial coordinates overlap by more than 50% with a YOLO-detected `abandon` region are discarded; and fifth, blocks overlapping by more than 50% with a YOLO-detected caption region are dropped to prevent the textual narrative from duplicating content already captured by the image crop. Surviving text blocks exceeding an empirical length of 40 characters are written to `extracted_content.json`.

4.2.4.3 3. Vision-Based Cross-Modal Post-Deduplication

A common challenge in document parsing is the duplication of captions that reside outside the spatial boundaries of the figure boxes. To remove this redundancy, a pretrained `DocTR` OCR engine executed in batches of size $B_{\text{ocr}} = 16$ processes all saved image crops to extract text visible within figures [35]. The extracted characters are mapped to a lowercase token vocabulary set defined as $\mathcal{V}_{\text{ocr}}$.

The pipeline runs a final validation pass over the textual blocks. For any short block whose length satisfies $4 \leqslant N_{\text{words}} \leqslant 22$, with a token vocabulary set $\mathcal{V}_{\text{block}}$, the system computes a Jaccard containment ratio against the OCR dictionary:

$$\text{Containment Ratio} = \frac{|\mathcal{V}_{\text{block}} \cap \mathcal{V}_{\text{ocr}}|}{|\mathcal{V}_{\text{block}}|}.$$

If this containment ratio meets or exceeds $T_{\text{intersection}} = 0.72$, the paragraph is flagged as a duplicate caption already captured by the visual layout crop and is dropped. Surviving block keys are reindexed contiguously per page to preserve the structural retrieval matrix.

4.2.5 Pipeline 1 Procedure

The complete procedural sequence for document layout analysis, visual consolidation, and cross-modal data filtration is formalized below.

Pseudocode, Pipeline 1: Image Extraction and Text Cleaning

1. **For each PDF** document d:
   (a) **For each page** p in d:
      i. Compute page layout statistics: dominant body style $(s_{\text{body}}, f_{\text{body}})$, median line height $\bar{h}$, dynamic gap $\tau_g = 1.4\,\bar{h}$
      ii. Filter lines: discard lines within top/bottom 2.5% margin
      iii. Group lines into blocks using vertical proximity, left-margin alignment, style consistency, sentence continuity, and figure-split rules → block list $\{b_1, \ldots, b_m\}$ with parallel bbox list
      iv. Rasterize p at 300 DPI → raster image R_p

v. Run DocLayout-YOLO on $R_p \rightarrow$ figure boxes F, caption boxes C, abandon boxes A

vi. Iteratively merge figure boxes in F within (H_m, V_m) margin; absorb caption boxes C into nearest figure box

vii. Crop each merged figure box (with 30 px padding) $\rightarrow$ save as `img_p{n}_{i}.png`

viii. Store A and C as pixel-to-PDF-coordinate scaled regions for page p

(b) **Cross-page stitching**: for each page boundary $(p, p{+}1)$, if last block of p is unfinished, prepend leading sentence of first non-empty block of $p{+}1$; replace consumed slot with ε

(c) **Build boilerplate sets**: collect exact and fuzzy block repeaters; collect 6-gram ($\geqslant 35\%$ pages) and 3-gram ($\geqslant 60\%$ pages) boilerplate n-grams

(d) **For each block** b_i on each page:

i. Skip if $b_i = \varepsilon$

ii. Strip boilerplate n-grams (6-gram pass, then 3-gram pass)

iii. Discard if TOC entry, structural repeater, $> 50\%$ overlap with abandon region, or $> 50\%$ overlap with caption region

iv. Emit to `extracted_content.json` if $|b_i| > 40$ characters

(e) Run DocTR OCR on all saved image crops (batches of 16) $\rightarrow$ `ocr_results.json`

(f) **OCR deduplication**: for each short block ($\leqslant 22$ words), compute Jaccard containment against all OCR token sets; discard if overlap $\geqslant 0.72$

(g) Renumber surviving block keys contiguously per page

(h) Free GPU and CPU memory and trigger explicit system garbage collection

For computational efficiency, the rasterization and layout detection steps are executed as a batched pass across groups of eight pages following the completion of text-block assembly, rather than strictly within the same iteration as each individual page. This grouping allows the DocLayout-YOLO inference calls to exploit GPU batch parallelism while preserving the per-page correctness of the overall procedure described above.

All thresholds and operational settings reported in this section were applied to the five-document evaluation corpus described in Section 5.2. The resulting extraction and filtering behaviour should therefore be interpreted as evidence for this heterogeneous French hydro-ecological document set, rather than as universally optimal settings for all technical PDF collections. Different document layouts, languages, scan quality, or visual conventions may require re-calibration of the layout, filtering, and deduplication thresholds.

4.3 Pipeline 2: Layout-Agnostic Image-Text Matching

Building on the spatially grounded text blocks and figure crops produced by Pipeline 1, the second core stage of our framework establishes a layout-agnostic multi-modal context retrieval engine, illustrated in Figure 4.5. Within unstructured technical environmental report corpora, images such as river hydro-geomorphology maps, sediment transport charts,

or species inventory matrices do not exist in isolation; they are deeply anchored to specific descriptive narratives, experimental parameters, or administrative discussions located in the surrounding text blocks. Pipeline 2 addresses the cross-modal indexing problem: given an extracted and isolated document image asset, the system identifies and retrieves the single most semantically relevant text block from the same source dossier. This retrieved block serves as a contextual grounding signal to guide downstream visual captioning and reduce structural hallucinations.

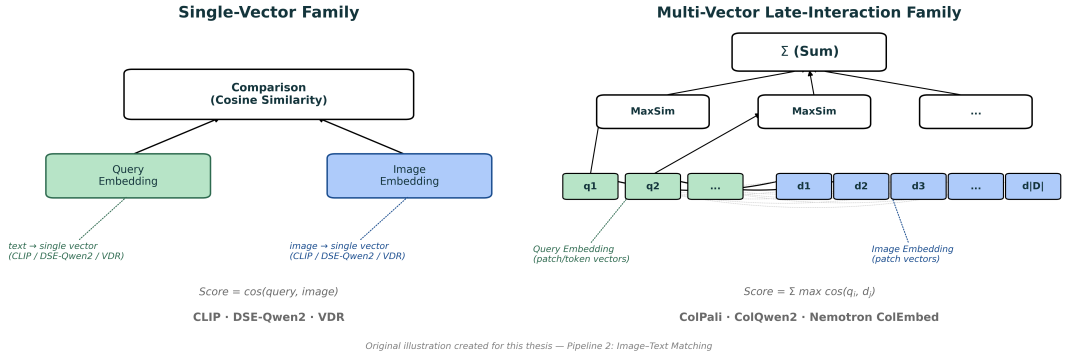


Figure 4.5: Structural architectural comparison between Single-Vector Dense Bi-Encoder Retrieval (Family 1) and Multi-Vector Late-Interaction Retrieval (Family 2) on multimodal document pages.

4.3.1 Family 1: Single-Vector Dense Bi-Encoder Retrieval

The classical baseline paradigm for cross-modal text matching frames retrieval as a dense single-vector intersection problem. Let $\mathcal{I}_i$ define a target extracted image asset and t_j define an isolated narrative text block. A dual-encoder architecture applies symmetric neural mapping paths to compress both modalities independently into a shared single-vector embedding space $\mathbb{R}^d$:

$$\mathbf{v}_i = f_\phi(\mathcal{I}_i), \quad \mathbf{h}_j = g_\psi(t_j), \quad \mathbf{v}_i, \mathbf{h}_j \in \mathbb{R}^d,$$

where f_ϕ and g_ψ represent the visual and textual encoding networks respectively. Relevance is evaluated monotonically using the cosine similarity of the isolated embedding vectors:

$$\text{Score}_{\text{Dense}}(\mathcal{I}_i, t_j) = \frac{\mathbf{v}_i \cdot \mathbf{h}_j}{\|\mathbf{v}_i\|_2 \|\mathbf{h}_j\|_2}.$$

Our methodology evaluates three primary architectures within this dense family:

- **Contrastive Language-Image Pre-Training (CLIP):** Evaluated using the `ViT-B/32` baseline framework [1]. Because it maps inputs to a single global representation using a general vision-language transformer trained on web image–text pairs, it serves as a conventional, layout-agnostic reference baseline. It is expected to have limited capacity to preserve fine-grained textual and spatial information contained in dense diagrams, structural charts, and technical tables. Its text tower is additionally bound by a hard 77-token input ceiling, meaning that long French narrative

passages are truncated before encoding. CLIP is therefore retained as an unmodified short-context contrastive baseline rather than as a fully configuration-matched competitive retriever. In particular, the remaining document-retrieval configurations process longer document representations and use the hybrid page-proximity reweighting term ($\beta = 0.02$), whereas CLIP does not. Its results should consequently be interpreted as the performance of a standard short-context CLIP configuration on paragraph-length technical document blocks. The present experiment does not disentangle the contribution of its 77-token context limit, the absence of proximity re-weighting, and single-vector global pooling to its observed retrieval deficit.

- **Dense Document Single-Vector Embedding (DSE-Qwen2):** Evaluated using the `MrLight/dse-qwen2-2b-mrl-v1` architecture[14]. This approach leverages a larger `Qwen2-VL-2B` backbone [16] to process the image, but still compresses the multi-modal hidden states into a single vector using the terminal token embedding. It incorporates Matryoshka Representation Learning (MRL) to allow flexible dimensional reduction while minimizing feature loss.

- **Visual Document Retriever (VDR)**: Evaluated using the `vidore/vdr-2b-multi-v1` configuration [5]. This model represents a continued fine-tuning of the DSE framework on a multilingual corpus of 500k query-image pairs. By downsampling visual sequence lengths by 70%, it lowers processing overhead on the HPC cluster, executing inference three times faster than its base network.

Although DSE-Qwen2 and VDR share a common Qwen2-VL backbone, each model's image preprocessing follows its own officially documented configuration rather than a unified scheme: DSE-Qwen2 applies a fixed-resolution resize, while VDR applies an aspect-ratio-preserving dynamic resize. Both configurations were retained as published to ensure each model is evaluated under the conditions it was originally trained and validated on.

While computationally efficient for matrix lookups, Family 1 suffers from a structural information bottleneck. Compressing complex page geometries, maps, or spatial diagrams into a single global vector discards local layout configurations, rendering it vulnerable to low-confidence alignments and narrow scoring margins when facing highly detailed environmental data.

4.3.2 Family 2: Multi-Vector Late-Interaction Retrieval

To address the single-vector bottleneck, the pipeline evaluates multi-vector late-interaction architectures. Instead of compressing an entire image or text block into a single vector, these models preserve the independent tokens or visual patches, allowing them to interact during the final matching stage via a maximum similarity operator. Our framework isolates three advanced models within this multi-vector family:

- **ColPali:** Built on a `PaliGemma-3B` backbone containing a `SigLIP-So400m` visual encoder [6]. It inputs images at a fixed 448×448 resolution, mapping document screenshots into a grid matrix of 1,024 independent patch vectors.

- **ColQwen2:** Converts images into flexible visual patch arrays using a dynamic resolution vision encoder to process variable aspect-ratio document components without distortion [6]. This approach generates up to 768 token patch representations, facilitating multilingual text recognition and instruction compliance.
- **Nemotron ColEmbed:** Built on a `Qwen3-VL-4B-Instruct` backbone, modified by converting the native causal self-attention mechanism into a bidirectional self-attention scheme to improve embedding quality for retrieval tasks [17]. It processes images through a dynamic-resolution vision encoder capped at 1,024 merged visual tokens per image (patch size 16, merge size 2), placing its representational ceiling on par with ColPali's fixed budget while retaining ColQwen2-style variable-aspect-ratio flexibility. The model is trained on a large-scale, diverse retrieval corpus incorporating hard-negative mining to sharpen late-interaction discrimination.

Let the query text block be represented as a matrix of token embeddings $\mathbf{Q} = [\mathbf{q}_1, \mathbf{q}_2, \ldots, \mathbf{q}_{|Q|}] \in \mathbb{R}^{|Q| \times d}$ and the document image asset be represented as a matrix of visual patch embeddings $\mathbf{D} = [\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_{|D|}] \in \mathbb{R}^{|D| \times d}$. The late-interaction mechanism calculates the maximum similarity between each query token and all document patches, summing the resulting scalars.

4.3.2.1 Design Choice 1: Mean-Normalized MaxSim Transformation

To address the token-count bias identified in Family 2, we introduce a Mean-Normalized MaxSim operator. The standard MaxSim scoring function sums scores across all text query tokens [15], which naturally biases the system toward longer target paragraphs, since longer text strings accumulate larger cumulative sums. Because technical report sections feature highly variable passage lengths, we instead divide the cumulative token score by the absolute query cardinality:

$$\text{Score}_{\text{Late}}(\mathcal{I}_i, t_j) = \frac{1}{|Q|} \sum_{k=1}^{|Q|} \max_{l=1}^{|D|} \left(\frac{\mathbf{q}_k \cdot \mathbf{d}_l}{\|\mathbf{q}_k\|_2 \|\mathbf{d}_l\|_2} \right).$$

This normalization improves alignment consistency across block lengths. We further implement a Block-Level Micro-Chunking strategy in place of full-page text pooling to prevent token dilution when scoring long text blocks against dense visual assets. Both adaptations were evaluated empirically against the single-vector family under identical conditions, with results reported in Section 5.6.

4.3.2.2 Design Choice 2: Micro-Chunking over Full Page Aggregation

Early architectural variants processed and matched text context at the full-page level by concatenating all textual sentences on a page into a single query string. Document parsing benchmarks revealed that full-page text pooling destabilizes the attention vectors of late-interaction models due to text token dilution. To resolve this, our methodology implements **Block-Level Micro-Chunking**. The image is matched independently against each separate text block isolated by Pipeline 1, and the specific text block that maximizes

the cross-modal similarity score is selected as the context match:

$$t^* = \arg\max_{t_j \in \mathcal{D}} \left(\text{Score}(\mathcal{I}_i, t_j)\right).$$

4.3.2.3 Design Choice 3: Hybrid Page-Proximity Re-Weighting

Raw semantic similarity scores alone can occasionally favor a text block that is topically related but physically distant from the source image within the document, which is undesirable for technical reports where figures are conventionally captioned or discussed in immediately adjacent text. To mitigate this, all models except the CLIP baseline apply a **hybrid re-ranking scheme** that combines min-max normalized semantic similarity with a linear page-distance penalty:

$$\text{Score}_{\text{Hybrid}}(\mathcal{I}_i, t_j) = \widehat{\text{Score}}(\mathcal{I}_i, t_j) - \beta \cdot |\, p(\mathcal{I}_i) - p(t_j)\,|,$$

where $\widehat{\text{Score}}(\mathcal{I}_i, t_j)$ denotes the min-max normalized similarity score across all candidate blocks for a given image, $p(\cdot)$ denotes the page number of the image or text block, and $\beta = 0.02$ is an empirically fixed distance-penalty coefficient. This soft constraint discourages implausible long-range matches without rigidly restricting the retriever to a fixed local window, preserving the ability to recover cross-page contextual matches when semantic alignment strongly warrants it.

4.3.3 The OCR Query Pollution Failure Axis

During the development of the multi-vector late-interaction track, an experimental variant explored injecting OCR-extracted text strings directly into the query matrix alongside the target text block. This approach was intended to provide a secondary alignment hint by matching text found inside the image with text in the report narrative. Environmental validation benchmarks revealed that this configuration triggers degradation in retrieval precision due to an OCR string-matching loop. When OCR tokens are appended to the query vector, late-interaction models exhibit a tendency to reduce reliance on multi-vector vision capabilities when interpreting layout features. Instead, the model's weights over-anchor onto literal text matching, forcing the visual patches to align with look-alike text patterns. This causes the retriever to overlook spatial, structural, and layout context, inducing alignment errors when facing matching data tables or figures with shared legend text. Consequently, OCR injection was removed from the retrieval pipeline. Clean, text-only block chunks are passed as queries to the late-interaction networks, prompting the encoders to rely on visual-linguistic grounding to map images to text.

4.3.4 Pipeline 2 Procedure

The complete procedural sequence for cross-modal context matching and evaluation is formalized below.

Pseudocode, Pipeline 2: Image-Text Matching

1. **Input:** Lossless extracted images $\{\mathcal{I}_1, \ldots, \mathcal{I}_M\}$, clean text blocks $\{t_1, \ldots, t_N\}$ for document d.

2. **For each candidate retrieval model $\mathcal{M}$:**
   (a) Parse and index all text blocks into vector formats: $\mathbf{T} = \{\mathcal{M}_T(t_j)\}_{j=1}^{N}$.
   (b) **For each image $\mathcal{I}_i$:**
      i. Process and compute the visual patch embedding matrix: $\mathbf{D}_i = \mathcal{M}_I(\mathcal{I}_i)$.
      ii. **For each indexed text block $\mathbf{T}_j$:**
         A. If $\mathcal{M}$ belongs to Family 1: compute s_{ij} using dense single-vector cosine similarity.
         B. If $\mathcal{M}$ belongs to Family 2: compute s_{ij} using Mean-Normalized MaxSim token-patch interaction.
         C. If $\mathcal{M} \neq$ CLIP: apply the hybrid page-proximity re-weighting adjustment to s_{ij}.
      iii. Rank the total text block collection descending by similarity score: $\mathcal{R} = \text{Sort}(\{s_{ij}\})$.
      iv. Retrieve the top-ranked text block (t^*) as the selected context match for image $\mathcal{I}_i$.
   (c) Evaluate model accuracy against ground-truth files using Recall@5, P@1, MRR, and nDCG@5.
3. Select and export the optimal cross-modal retriever configuration to drive the Pipeline 3 captioning stage.

The context blocks retrieved by this pipeline provide the grounding signal required for the visual captioning stage described next, which combines this local textual context with the raw pixel content of each figure to generate factually anchored descriptions.

4.4 Pipeline 3: Grounded Visual Captioning

Following layout segmentation and cross-modal context retrieval, the pipeline introduces an intelligent, context-grounded visual captioning layer. Within technical engineering reports, generic visual descriptors are structurally insufficient; a caption must transform raw pixel observations into domain-specific, verifiable knowledge elements. Pipeline 3 establishes a multi-modal generation framework designed to balance descriptive novelty with strict factual alignment, utilizing a two-phase prompting strategy and cross-modal verification safeguards to mitigate visual hallucinations, as illustrated in Figure 4.6.

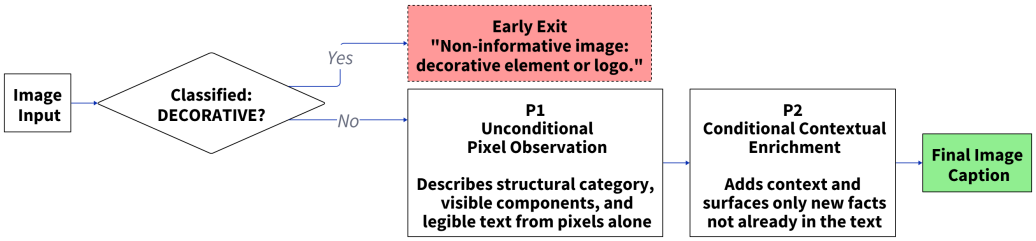


Figure 4.6: Structural workflow of the two-phase prompting architecture, demonstrating how linguistic confirmation biases are neutralized by separating visual inventory compilation from contextual enrichment.

4.4.1 Low-Resolution VLM Limitations

Initial exploratory phases evaluated standard, zero-shot vision-language architectures, such as the BLIP-1 [20] and BLIP-2 [2] models, to establish a baseline for information discovery. When tested against highly granular hydromorphological maps, site photographs, and multi-layered engineering diagrams, these models frequently collapsed into generic, high-level, or geographically incorrect descriptions. Figure 4.7 illustrates this limitation using a set of four river-restoration photographs. BLIP-1 generated the generic description *"four pictures of a river with a few different types of water"*, whereas BLIP-2 produced the geographically inaccurate output *"a map of the st john's watershed"*. This diagnostic failure stems from two architectural limits: a frozen visual encoder that collapses spatial token details, and an absence of localized optical character recognition (OCR) grounding.



Input comprising four river-restoration photographs.

BLIP-1 output

"four pictures of a river with a few different types of water"

BLIP-2 output

"a map of the st john's watershed"

Figure 4.7: Zero-shot BLIP baseline outputs for four river-restoration photographs. BLIP-1 produces a generic visual description, while BLIP-2 incorrectly classifies the photographic input as a map and assigns an unsupported geographical identity.

The baseline models also lacked a reliable mechanism for distinguishing informative technical visuals from non-informative layout elements such as institutional logos, page banners, and decorative graphics. Together, these limitations motivated the transition toward larger, instruction-following vision-language models capable of preserving higher-resolution visual information and incorporating retrieved textual context during caption generation. The subsequent captioning pipeline therefore evaluates models with stronger document-understanding, OCR, and spatial-reasoning capabilities rather than relying on fixed-resolution zero-shot baselines. The model selection and ablation strategy are detailed in the following section.

4.4.2 VLM Selection and Truncation Gates

To address the visual resolution bottleneck, the architecture integrates advanced conversational vision-language models capable of dynamic patch allocation. Candidate models evaluated within this framework: Pixtral-12B[27], Mistral-Small-3.1-24B[36], Qwen3-VL-30B[37], NVILA-Lite-8B[38], Llama-3.2-Vision-11B[39], InternVL2.5-8B[40], and Qwen3.6-27B[41] — are selected for evaluation according to four structural characteristics:

1. **Native Resolution Agnostic Routing:** The model features a native vision encoder capable of decomposing variable aspect-ratio images into flexible sub-patch tokens, avoiding the downsampling distortions that corrupt low-resolution architectures.

2. **Instruction-Guided Sequence Alignment:** It functions as a conversational auto-regressive model, allowing precise domain personas and structural constraint logic to be enforced directly within the attention layers.

3. **Programmatic Early-Exit Truncation:** To protect downstream knowledge graph extraction passes from processing non-informative corporate symbols, the prompt instructs the model to categorize non-informative layout elements under a strict `DECORATIVE` token wrapper. When a logo triggers this flag, the generator limits its attention weights and forces an immediate sequence termination, outputting a standardized string: *"Non-informative image: decorative element or logo"*.

4. **Document-Heavy Performance profiles:** The model demonstrates competitive document-level precision scores on established public benchmarks such as DocVQA[42] and MathVista[43], reflecting its capacity to parse dense, text-rich visual planes.

4.4.3 Prompt Topology and Alignment Bias

The structural accuracy of multi-modal descriptions is highly sensitive to the topological design of the instruction prompt. Our methodology systematically developed and evaluated three distinct prompt layouts to isolate and neutralize hallucination vectors. The prompt templates were written in French when operating on the French environmental report corpus. The representative templates presented below preserve the experimental instruction order, role assignment, required output structure, and principal grounding constraints. Minor wording refinements were made during development.

4.4.3.1 1. Prompt 1: Structured Analyst Baseline

The initial design positioned the VLM as a neutral scientific analyst reviewing figures extracted from French environmental and hydrology reports. The prompt enforced a strict taxonomic filter, requiring the model to classify the asset as `[MAP, GRAPH, DIAGRAM, PHOTO, TABLE, DECORATIVE]`. If classified as non-informative, the model was required to return only a fixed decorative-element message. Otherwise, it produced a structured description covering: (a) the primary subject matter; (b) visible data, measurements, labels, and annotations; (c) spatial relationships, trends, or processes; and (d) the meaning of embedded French labels. The prompt explicitly required factual French-language output and prohibited speculation beyond the visible image content. While Prompt 1 achieved strong descriptive accuracy on isolated geographic charts, it demonstrated an unstable grounding profile when handling complex, long-tail data variations. **Representative**

Prompt 1 Template.

1. **Role:** Expert scientific analyst reviewing figures extracted from French environmental and hydrology reports.

2. **Classification:** Identify the image type as `MAP`, `GRAPH`, `DIAGRAM`, `PHOTO`, `TABLE`, or `DECORATIVE`.

3. **Early exit:** If the input is decorative or a logo, return only: *"Non-informative image: decorative element or logo."*
4. **Description:** For all other image types, report the visible subject, data and annotations, spatial relationships or processes, and the meaning of French text or labels.
5. **Constraints:** Be precise and factual, do not speculate beyond visible evidence, and write in French.

4.4.3.2 2. Prompt 2: Topological Language Confirmation Bias

To incorporate specialized domain vocabulary, Prompt 2 adopted a targeted persona focused on forensic hydro-geomorphology. This topology reversed the data flow by injecting the full multi-modal context, consisting of late-interaction matched text blocks paired with raw image OCR strings, prior to the image observation section. The response was structured across four distinct sections: `CLASSIFICATION`, `ANCRAGE & RÉALITÉ`, `INDICATEURS TECHNIQUES`, and `DIAGNOSTIC EXPERT`. Experimental evaluation demonstrated that Prompt 2 generated higher error rates across factual precision metrics. Diagnostic review identified a critical failure mechanism termed Topological Language Confirmation Bias. Pre-loading dense linguistic context before image analysis caused the model to over-anchor onto textual terms. The model then used visual information to confirm text priors rather than reporting the actual visible content. For example, when retrieved text discussed pond restoration techniques, the model could report active water features within dry geomorphological cross-sections. Furthermore, forcing the model to complete highly specialized technical slots created pressure to generate fabricated observations when the source figure lacked relevant evidence. **Representative Prompt 2 Template.**

1. **Role:** *"Tu es un EXPERT AUDITEUR en hydro-géomorphologie fluviale. Ton rôle est d'extraire des faits physiques atomiques pour un graphe de connaissances."*
2. **Inputs provided before visual diagnosis:**
   - Report context: `{image_context}`
   - Image OCR text: `{image_ocr}`
3. **Required response sections:**
   (a) `CLASSIFICATION`: image type and decorative-image early exit.
   (b) `ANCRAGE & RÉALITÉ`: technical object identification and verification of OCR dates, numbers, and names against report context.
   (c) `INDICATEURS TECHNIQUES`: visible sediment state, physical dynamics, and hydraulic or biological state.
   (d) `DIAGNOSTIC EXPERT`: functional analysis and any discordance between the context and visible evidence.
4. **Constraints:** Produce factual, short, and technical observations; avoid evaluative adjectives and avoid paraphrasing the source report.

4.4.3.3 3. Prompt 3: Two-Phase Pixel-First Architecture

To reduce language confirmation bias, the final prompt topology was designed from first principles, combining three distinct theoretical frameworks on hallucination mitigation:

- **Visual Description Grounded Decoding (VDGD):** Enforces a strict visual isolation boundary [7]. The prompt splits generation into two sequential phases. Phase 1 bars all external context text, forcing the model to compile an assumption-free, feature-level inventory of raw pixels before performing domain interpretation.
- **Multi-Modal Mutual-Information Decoding (M3ID):** Restricts the sequence length of the subsequent interpretation phase [18]. Phase 2 is limited to short factual updates to reduce opportunities for unsupported elaboration.
- **Black-Box Visual Prompt Engineering:** Motivates the use of specific visual indicators and explicit grounding constraints [19]. The final instruction directs attention toward observable geomorphological features without permitting unsupported inferences.

The execution sequence of Prompt 3 is formalized as follows.

Phase 1: Unconditional Pixel Observation. The model constructs an empirical visual index using only the image pixels. It identifies the structural category, lists visible components, and transcribes legible alphanumeric strings. If an element falls below the visibility limit of the raster layer, the model is restricted from making assumptions and outputs: *"non observable à cette échelle."*

Phase 2: Conditional Contextual Enrichment. The model is subsequently provided with retrieved local context and image OCR. It identifies one or two physical facts that are visible in Phase 1 but absent from the report context. If no novel visual fact can be verified, the model outputs: *"Aucun enrichissement détecté."* The final grounding rule prohibits Phase 2 from introducing an element that was not already identified in Phase 1.

Representative Prompt 3 Template.

1. **General instruction:** *"Tu travailles sur des documents environnementaux français portant sur la restauration de cours d'eau, le Rhin et l'hydromorphologie."*
2. **Phase 1, image-only description:**
   - Describe only what is visible in the pixels.
   - Classify the image as `PHOTO`, `MAP`, `GRAPH`, `DIAGRAM`, or `TABLE`.
   - Identify the visible subject, physical elements, and readable text.
   - For logos or decorative elements, return only: *"Image non informative : élément décoratif ou logo."*
   - For elements that cannot be identified reliably, return: *"non observable à cette échelle."*
   - Do not fill gaps using assumptions.
3. **Phase 2, contextual comparison:**

- Report context: `{image_context}`
- OCR text: `{image_ocr}`
- Identify one or two physical observations found in Phase 1 but absent from the retrieved context.
- If no new fact is visible, return: *"Aucun enrichissement détecté."*
- If the image contradicts the context, state the discordance in one sentence.

4. **Final constraint:** Phase 2 may mention only elements already declared in Phase 1; no speculative addition is permitted.

4.4.4 MapReduce Context Summarization

Phase 2 uses both document-level context and image-specific retrieved text blocks. To extract the global document context without exceeding the attention windows of our models, the pipeline implements a multi-stage MapReduce Text Summarization workflow driven by an intermediate processing node using `Qwen2-VL` [16] as the underlying summarization model, as illustrated in Figure 4.8.

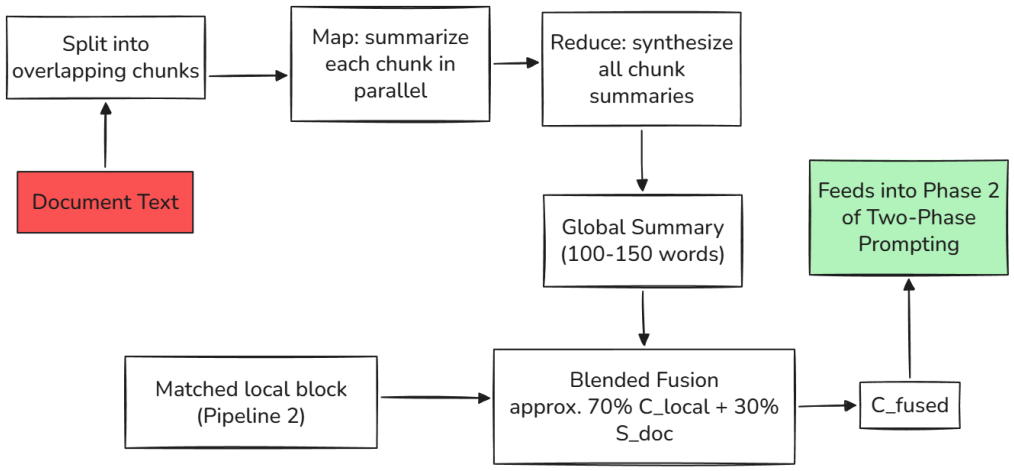


Figure 4.8: The MapReduce abstractive summarization framework, demonstrating overlapping token chunking followed by global context reduction.

Let the complete unstructured document text string be represented as an array of words $\mathcal{T}$. The MapReduce sequence executes across two distinct processing phases:

1. **The Map Phase:** The text array $\mathcal{T}$ is split into consecutive token blocks $x_k \in \mathbb{R}^L$ of length $L = 2500$ words. To prevent semantic sentence boundary loss along the slice coordinates, an overlapping window buffer of $O = 150$ words is maintained between adjacent chunks. Each isolated block is processed in parallel using a localized factual extraction prompt:

$$s_k = \text{LLM}_{\text{Map}}(x_k), \quad \forall k \in \{1, \ldots, K\},$$

which condenses each text chunk into a dense, 60-to-80-word factual summary segment s_k.

2. **The Reduce Phase:** The intermediate summary tokens are concatenated into a unified vector stream and passed through a final synthesis layer to resolve cross-chunk redundancies and assemble the master abstractive document summary S_{doc}:

$$S_{\text{doc}} = \text{LLM}_{\text{Reduce}}\left([s_1 \parallel s_2 \parallel \cdots \parallel s_K]\right).$$

The final output is limited to 100–150 words and summarizes the primary environmental objectives, locations, and engineering actions.

During initial benchmarking, a native French transformer architecture, `BARThez`, was evaluated as a localized alternative for the reduction pass [44]. It contains a bidirectional BART encoder-decoder topology pretrained on a large corpus of native French text. While computationally efficient during inference, its static 1,024-token context window required aggressive text chunking, resulting in lower semantic coherence on specialized hydrological report data compared to the larger multi-modal network configuration.

4.4.5 Context Fusion Strategies

Our framework evaluated three context fusion strategies to optimize the information flow entering Phase 2:

- **Strategy 1 (Local Only):** Passes only the localized text blocks C_{local} retrieved via Pipeline 2, omitting macro-thematic guidelines.
- **Strategy 2 (Parallel Unblended):** Appends the document summary S_{doc} and localized blocks C_{local} as separate, unblended prompt sections, which can cause mid-size models to generate redundant text due to attention dispersion across unblended context.
- **Strategy 3 (Blended Pre-Fusion):** Rather than computing a literal numerical interpolation, which is not well-defined for natural language text, an intermediate language model is instructed to synthesize the two context sources into a single, coherent passage that allocates approximately 70% of its content to the localized text blocks C_{local} and up to 30% to the document summary S_{doc}, denoted informally as $C_{\text{fused}} \approx 0.70 \cdot C_{\text{local}} \oplus 0.30 \cdot S_{\text{doc}}$, where $\oplus$ denotes instruction-guided content fusion rather than numeric addition. The document summary is restricted to supplying terminology and thematic framing, while the localized blocks remain the primary factual source. Under this configuration, local blocks provide the main factual context, while the document summary provides broader thematic context when local retrieval is incomplete. Strategy 3 is adopted as the standard for pipeline deployment. Neither the MapReduce-generated document summaries (S_{doc}, Section 4.4.4) nor the weighted-context fusion outputs used as Phase 2 grounding signals were reviewed or corrected by a domain expert; any systematic bias or error in the automatically generated summaries propagates directly into the fused context and, downstream, into caption grounding scores (Section 5.7.4).

4.4.6 Prompt-Engineering Interventions

Following the selection of the base configuration described in the experimental results, six inference-time prompt variants are evaluated using the configuration selected for downstream caption generation. Model weights, retrieved context, the evaluation corpus, and the downstream evaluation procedure remain unchanged; the variants therefore isolate alternative instruction designs rather than model architectures.

These theory-informed interventions were implemented for this thesis. Their labels denote experimental configurations rather than direct replications of identically named published methods, and are motivated by work on pixel-grounded description, prompt-based hallucination mitigation, and multimodal grounding constraints [7, 18, 19].

- **A5-P0: Baseline.** Uses the selected two-phase pixel-first prompt and blended context without an additional mitigation instruction.
- **A5-P1: Visual Primitive Guardrail.** Restricts the initial description to directly observable visual primitives, such as objects, colours, shapes, spatial relations, and readable labels, before allowing contextual interpretation. This tests whether a stronger observation-first constraint improves visual grounding [7, 19].
- **A5-P2: Verification and Refusal Gate.** Requires the model to withhold a claim when the available image evidence is insufficient to support it. This tests whether explicit uncertainty and refusal constraints reduce unsupported contextual additions [7, 19].
- **A5-P3: Counterfactual Contrast.** Requires the model to identify relevant elements suggested by the retrieved context that are not visually supported by the image before producing the final description. This negative-evidence check, termed *Bilan d'Absence* in the prompt, is intended to reduce confirmation of context-derived assumptions. It is inspired by contrastive visual-grounding principles, but is not a direct replication of a published method [7, 18].
- **A5-P4: Internal Self-Correction.** Requires an initial candidate description to be reviewed against the visible image evidence before the final response is emitted. This configuration is conceptually related to verification-and-revision approaches such as Chain-of-Verification [45], while remaining a lightweight single-model prompt procedure.
- **A5-P5: Domain Expert Persona.** Assigns a specialist hydro-geomorphology role and domain vocabulary to the model. This tests whether stronger task framing improves the usefulness of technical descriptions, while also testing whether specialised framing can encourage over-interpretation beyond visible evidence.

The complete prompt text is retained in the accompanying implementation repository. Ablation 5 evaluates these six configurations under the common model and context setting described above; its results are reported in Section 5.7.5.

4.4.7 Contrastive Decoding Interventions

Ablation 6 evaluates two training-free contrastive decoding methods that modify token selection during generation rather than changing the prompt or fine-tuning the captioning model. Both methods combine the logits from a standard generation pass with logits from a deliberately perturbed pass, thereby reducing the influence of predictions that remain likely when the visual or instructional evidence is weakened.

4.4.7.1 Instruction Contrastive Decoding

Instruction Contrastive Decoding (ICD) contrasts next-token logits from the original instruction with those from a contrasting instruction intended to expose language-prior-driven generation [46]. The image input remains fixed while the instruction is perturbed. Let z_t^+ denote the logits generated under the original instruction and z_t^- the logits generated under the contrasting instruction at decoding step t. The adjusted distribution is obtained from:

$$p_{\mathrm{ICD}}(y_t \mid y_{<t}) = \mathrm{softmax}\big(z_t^+ + \alpha\,(z_t^+ - z_t^-)\big)\,,$$

where α controls the contrastive strength. An adaptive plausibility mask excludes tokens whose original logits fall below a threshold relative to the highest-scoring token. Larger α values increase the instruction-contrast signal, but may also suppress useful descriptive content.

In the present implementation, the original image remains unchanged in both branches. The primary branch receives the full two-phase, retrieval-grounded prompt, whereas the contrastive branch receives a short generic instruction requesting a simple image description. This disturbance instruction removes the domain persona, phase structure, retrieved context, and grounding constraints. ICD is evaluated only through the contrastive strength parameter, using $\alpha = 0.5$ and $\alpha = 1.0$.

4.4.7.2 Visual Contrastive Decoding

Visual Contrastive Decoding (VCD) contrasts token predictions conditioned on the original image with predictions conditioned on a visually perturbed image [47]. Let z_t^{orig} be the logits obtained from the original image and z_t^{pert} those obtained from the perturbed image. The contrastive distribution is:

$$p_{\mathrm{VCD}}(y_t \mid y_{<t}) = \mathrm{softmax}\Big(z_t^{\mathrm{orig}} + \alpha\,\Big(z_t^{\mathrm{orig}} - z_t^{\mathrm{pert}}\Big)\Big)\,.$$

In this study, the perturbed branch applies additive Gaussian diffusion noise to the processed image tensor using a 1,000-step sigmoid beta schedule:

$$x_t = \sqrt{\bar{\alpha}_t}\,x + \sqrt{1-\bar{\alpha}_t}\,\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I),$$

where $\bar{\alpha}_t$ is the cumulative product of the noise schedule. Both VCD configurations use $t = 500$; mild and strong conditions differ only in contrastive strength ($\alpha = 0.5$ and $\alpha = 1.0$, respectively). The text prompt, retrieved context, and tokenized input structure remain identical between the clean and perturbed branches. The same adaptive plausibility constraint is applied, while required phase-header and early-exit tokens are protected

from contrastive modification. VCD is intended to favour tokens whose likelihood depends on the original visual evidence rather than on language priors that persist when the visual signal is degraded.

Both ICD and VCD were evaluated at mild ($\alpha = 0.5$) and strong ($\alpha = 1.0$) contrastive strengths. These settings test the trade-off between suppressing unsupported generation and retaining sufficient descriptive vocabulary for technical image captions. The results are reported in Section 5.7.6.

4.4.8 Pipeline 3 Procedure

The complete procedural sequence for context summarization, two-phase visual description generation, and cross-modal token checking is formalized below.

Pseudocode, Pipeline 3: Context-Grounded Image Captioning

1. **Input:** Consolidated image crops $\{\mathcal{I}_1, \ldots, \mathcal{I}_M\}$ and text blocks $\{t_1, \ldots, t_N\}$ from Pipeline 1.
2. Execute the MapReduce summarization engine over $\{t_j\}$ using overlapping token chunks to generate the global abstractive summary S_{doc}.
3. **For each image** $\mathcal{I}_i$:
   (a) Extract the top matching local text blocks via Pipeline 2 to obtain C_{local}.
   (b) Construct the unified context string by instructing an intermediate language model to fuse C_{local} and S_{doc} into a single passage with an approximate 70/30 content emphasis, denoted $C_{\text{fused}} \approx 0.70 \cdot C_{\text{local}} \oplus 0.30 \cdot S_{\text{doc}}$.
   (c) Build Prompt 3 by configuring the forensic persona parameters and mapping the visual indicator vocabulary.
   (d) Feed the image asset and prompt sequence to the target conversational vision-language model:
   $$\hat{c}_i = \text{VLM}_{\text{Decoder}}\left(\mathcal{I}_i, \text{Prompt } 3(C_{\text{fused}})\right).$$
   (e) If $\hat{c}_i$ matches the `DECORATIVE` early-exit signature string, immediately strip the asset from downstream extraction tasks to protect downstream data integrity.
4. Deconstruct surviving output strings into independent atomic sub-claims using an isolated verification node.
5. Compute cross-modal validation metrics consisting of the Visual Grounding Rate, the VQA Atomic Score, and sentence-level NLI Novelty.

With grounded, factually anchored captions now available for every non-decorative figure, the pipeline proceeds to unify these image-derived observations with the textual narrative stream into a single, schema-compliant knowledge graph, as described in Pipeline 4.

The captioning models evaluated in this work are used in zero-shot/prompted configuration only. Supervised fine-tuning to further reduce hallucination would require an expert-annotated corpus of 100–200 image–caption pairs, which falls outside the time and annotation-resourcing scope of this analysis.

4.5 Pipeline 4: Cross-Modal Semantic Integration

Following text extraction and grounded visual caption generation, the pipeline introduces a cross-modal knowledge graph population and semantic integration layer. Unstructured report narratives and verified visual descriptions capture complementary dimensions of the underlying environmental interventions. If processed in isolation, critical topological linkages between textual facts and visual evidence remain disconnected. Pipeline 4 maps textual and image-derived observations into a schema-compliant Resource Description Framework (RDF) knowledge graph structured by the `TetraOnto` domain ontology. This design allows image-derived physical, spatial, and temporal attributes to be linked with text-derived entities.

4.5.1 Limits of Direct Generative Serialization

Initial graph construction strategies evaluated a direct serialization paradigm, instructing the generative model to output graph data directly in native W3C Turtle syntax [28]. Empirical testing revealed a structural vulnerability in this approach due to generative syntax breakage constraints.

Large language models can produce malformed raw RDF syntax when asked to generate code containing escaping rules, namespace declarations, and nested structures. When processing French environmental data, the models frequently output unescaped apostrophes, unencoded special symbols, or raw mathematical brackets within string literals. These syntax anomalies trigger immediate parsing crashes in downstream RDF parsers such as `rdflib` [48], leading to malformed graph files and systemic file corruption.

To reduce serialization failures, the pipeline separates generative extraction from deterministic RDF serialization. The task is split into two decoupled modules: first, an LLM-driven extraction module restricted to structural data formatting, and second, a programmatic conversion module managed by a deterministic Python serialization script. By restricting the language model to emitting standard web-compliant JSON arrays containing flat `subject`, `subject_type`, `predicate`, and `object` text slots, syntax parsing bugs are eliminated. The downstream Python script ingests clean string tuples, sanitizes special characters, maps unicode variations, and executes programmatic triple creation using safe native API methods. If a document paragraph triggers a malformed JSON string, the handler attempts bounded salvage repair (truncating to the last well-formed object before parse failure) and, failing that, drops the block from the active pass and logs the anomaly, protecting the overall execution run from file corruption.

4.5.2 Schema Ingestion and Prompt Conditioning

A major challenge in deploying large language models for automated knowledge graph population is preventing schema-compliance drift. If an open-source model is prompted to extract relation triples without strict structural boundaries, it can invent subjective classes and arbitrary predicates that violate formal ontological schema constraints. To enforce compliance with our custom domain schema without exceeding model context windows, our pipeline introduces a specialized Ontological Schema Ingestion and Compression Pre-Pass.

4.5.3 The TetraOnto Domain Schema

The semantic target utilized across all extraction and graph integration stages in this work is the `TetraOnto` domain ontology, established in prior work by Ghazouani, Giustozzi, and Le Ber [3, 24] for hydro-ecosystem restoration. `TetraOnto` models the domain conceptual space through a formal hierarchy centered around seven primary ontological classes:

1. `restorationMeasure`: represents the physical ecological interventions, such as river reconnection, bank restoration, weir removal, and fishway installation;
2. `structure`: models artificial and natural physical obstacles or technical elements, including dams, weirs, sills, and step-pool fish passes;
3. `waterBody`: designates the aquatic environments affected by restoration operations, including rivers, channels, and wetlands;
4. `geographicZone`: captures the administrative and spatial footprints of the intervention sites;
5. `migratorySpecie`: defines biological target species affected by river continuity alterations;
6. `projectOwner` and `mainContractor`: identify the administrative and technical organizations managing the restoration campaigns.

These core entities are interconnected via domain-specific OWL object properties, notably `realizedIn` (linking an intervention to a geographic zone), `associatedTo` (associating an intervention with a physical structure), `isLocatedOn` (positioning a barrier along a designated water body), and `isManagedBy` (mapping operations to stakeholders). Quantitative and temporal metadata are formalized using datatype properties, such as `startsAt`, `endsAt`, `hasCost`, `hasHeight`, and `isImpassable`.

In parallel with this compression stage, a structural summary of the ontology was generated in order to characterize the schema before prompt injection. This analysis confirmed that the source file is a pure TBox resource, with no named individuals, and that its organization is strongly centered around hierarchical class design and restriction-based modeling. Such observations are methodologically important because they support the use of ontology-guided prompt conditioning as a schema-control mechanism rather than an instance-level enrichment step.

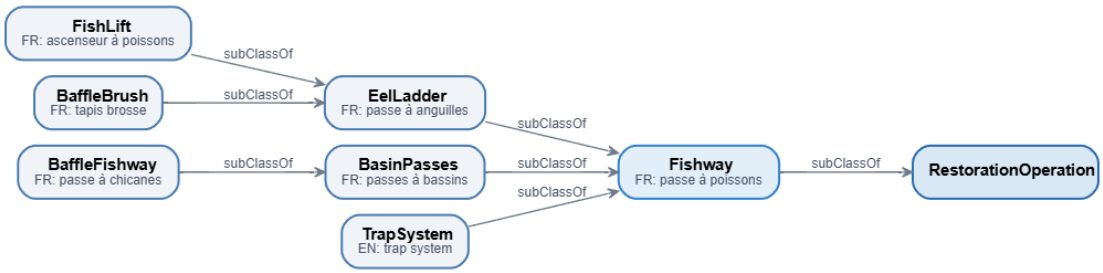


Figure 4.9: Restoration-operation class hierarchy in the `TetraOnto` ontology.

A compact summary of the extracted ontology structure is given below:

- Classes: 92
- Properties: 48
- Subclass links: 80
- OWL restrictions: 93
- Named individuals: 0
- Main class hub: `restorationMeasure`

It should be noted that `TetraOnto` was developed and refined primarily against the text-only corpus studied in prior work; it was not re-audited or extended against the specific visual and thematic content of the multi-modal document set used in this thesis. Consequently, some visually derived entities and relations (e.g., structural or monitoring attributes visible only in images) may not map cleanly onto existing ontology classes, which is a contributing factor to the residual schema-conformance gap observed in Section 5.8.2.

These statistics indicate that the ontology is structurally rich despite its moderate size, with a strong emphasis on class specialization and logical restriction patterns. In particular, the high number of subclass links and OWL restrictions suggests that the schema is not merely a flat vocabulary, but a formally organized conceptual backbone. The identification of `restorationMeasure` as the principal hub further suggests that restoration-oriented concepts serve as one of the central organizing axes of the ontology.

In addition to class and property compression, the schema-compliance mechanism is enforced at two independent points in the pipeline: (i) at generation time, via the injected compact schema, and (ii) at serialization time, via a deterministic predicate and class whitelist parsed directly from the same ontology file (Section 4.5.5). This dual enforcement prevents non-conformant predicates and classes from entering the final graph as ontology relations.

By embedding this structural lookup schema directly inside the prompt context, the model's self-attention heads are guided to anchor onto authorized properties. However, prompt conditioning alone does not guarantee compliance; empirical testing (Section 4.5.8) showed that a meaningful fraction of generated predicates still fall outside the authorized vocabulary even with the full schema in context, which is why the whitelist-and-demotion mechanism described below is applied as a mandatory downstream safeguard rather than a redundant check.

4.5.4 Two-Pass Extraction and Coreference Resolution

Extraction proceeds in two sequential passes per source document: a text pass over paragraph-level content blocks, followed by an image pass over the verified captions produced by Pipeline 3. The image pass inherits the running entity registry built during the text pass, a bounded list (capped at 40 most-recently-seen entities to control prompt token growth) of entity labels already identified in the document. This registry is re-injected

into the prompt for every subsequent call, together with an explicit coreference instruction directing the model to reuse the exact name of a registry entity whenever the input text refers to it generically, implicitly, or via a pronoun, rather than creating a new entity. This mechanism is the primary means by which image captions are linked to entities already established from text, prior to any downstream graph-level merging.

To manage throughput, extraction calls are executed in batches (four content blocks per batch for text, two per batch for images) using padded, left-truncated batched generation. If a batch triggers a CUDA out-of-memory condition, the pipeline automatically falls back to sequential, single-item processing for that batch rather than failing the run. Content blocks exceeding a fixed word threshold are additionally split into overlapping chunks (2500 words per chunk, 150-word overlap) prior to extraction, with partial-chunk outputs merged and deduplicated on the (subject, predicate, object) key before being passed to the RDF serialization stage.

4.5.5 Ontology Conventions and Validation

The serialization module maps validated JSON triples onto strict W3C-aligned modeling rules inside the database:

- **Individual Declarations:** Every unique subject, and each object classified as an entity reference, is declared as an `owl:NamedIndividual`; literal objects are serialized as RDF literals. [29].
- **Class-Type Resolution:** Each subject's model-predicted `subject_type` is checked against the whitelist of valid ontology classes parsed from the compact schema. If the predicted type is not a recognized class, a secondary fallback attempts to resolve it via a table of SKOS `altLabel` synonyms mapped to their canonical class URIs, built by re-parsing the full, uncompressed `TetraOnto.ttl` file directly (rather than the compact summary), since it requires the complete set of alias annotations rather than the token-optimized subset injected into the prompt; if no match is found, the individual is left untyped rather than assigned an invented class.
- **Predicate Whitelisting and Demotion:** Every predicate is checked against the ontology's authorized property list. Predicates outside this whitelist are not discarded outright; instead, the fact is preserved as a `skos:note` literal attached to the subject (recording the original predicate name and object value), demoting it from a structural graph edge to an auditable annotation. This design choice keeps the visualized graph schema-conformant while retaining a full audit trail of information the model attempted to extract but that fell outside the authorized vocabulary, supporting the Schema Conformance Rate metric defined in Section 4.5.9.
- **Property Classification:** Retained (whitelisted) predicates are classified by checking their range profile. Links connecting two independent graph nodes are assigned as an `owl:ObjectProperty`, while links connecting an entity node to a literal value are mapped as an `owl:DatatypeProperty` [29]. An object value is treated as an entity reference (rather than a literal) if it is short (six words or fewer) and not purely numeric; longer or numeric values are serialized as literals instead.

- **Artifact Filtering:** Two lightweight heuristic filters run prior to triple creation: numeric-literal detection prevents raw measurements or counts from being reified as entities, and a section-number pattern filter discards table-of-contents-style artifacts (e.g., `"1.2.3"`) that occasionally leak into subject or object fields from PDF layout noise. A further predicate-specific filter rejects date-typed predicates whose object value is actually a bare page number, redirecting the fact to a `skos:note` annotation instead of a spurious date edge.

- **Multi-Modal Provenance Tracking:** Each generated entity node is annotated with provenance metadata: `:dataSource` (the source document title) and `:extractedFrom` (`"Text"` or `"Image"`, with image-sourced facts additionally tagged `"VQA-verified caption"` to reflect their Pipeline 3 provenance).

- **Spatial Coordinate Mapping:** Document coordinates are preserved by binding each fact to an `xsd:integer` data field labeled `:hasPageNumber`, mapping the database node back to its physical page location. This field is stored as node metadata rather than embedded in the entity URI itself (see Section 4.5.6 for the URI construction scheme).

- **Ontology Architecture Integration:** Every compiled file is prefixed with the TetraOnto namespace declarations and an `owl:Ontology` header, alongside SKOS concept labeling (`skos:prefLabel` for canonical entity names) [30].

4.5.6 Cross-Modal Entity Resolution

Unlike a coordinate-keyed URI scheme, entity identity in this pipeline is resolved through label-based matching rather than shared page coordinates, since two mentions of the same real-world entity frequently appear on different pages or under paraphrased names. Each entity's URI is instead derived deterministically from its cleaned surface label via Unicode normalization, accent stripping, and title-casing (`normalize_uri_fragment`), with page number and provenance retained separately as node attributes rather than encoded into the identifier.

Cross-modal fusion of text-derived and image-derived individuals into shared graph nodes is achieved through a three-stage resolution process applied after the text and image graphs are merged:

1. **Exact label normalization merge:** Entities sharing an identical `skos:prefLabel` after lowercasing, whitespace normalization, and stripping of French definite/indefinite articles (le, la, les, l', un, une, des, du, de la) are folded into a single canonical node, chosen as the URI with the highest existing triple count.

2. **LLM-mediated alias resolution:** The full list of distinct entity labels present in the merged graph is passed to the language model with a dedicated coreference-resolution prompt, which is asked to identify pairs where one label is a generic or pronominal reference (e.g., "le site", "cet ouvrage") and the other is the specific proper name of the same real-world object within the same document. Confirmed pairs are folded into the canonical (proper-name) node.

3. **Similarity-based alias resolution:** Remaining near-duplicate labels are compared using normalized string similarity (Python's `difflib.SequenceMatcher`, threshold $\geqslant 0.92$), with a numeric-suffix guard preventing entities differentiated only by an index (e.g., "Tronçon 1" vs. "Tronçon 2") from being incorrectly merged.

The exact-label merge step is re-run after each of the two alias-resolution passes to consolidate any newly-identical labels produced by the fold. After the full text and image passes complete, the pipeline additionally logs three diagnostic counts used for downstream evaluation (Section 4.5.9): the number of text-only entities, the number of entities newly introduced by the image pass, and the number of entities present in both modalities (cross-modal linked entities), computed via set intersection over `owl:NamedIndividual` instances in the respective text and image graphs prior to merging.

4.5.7 Model Fine-Tuning for Predicate Conformance

To reduce the rate at which the generative extraction model emits predicates outside the authorized ontology vocabulary, a parameter-efficient fine-tuning procedure was applied to the base extraction model (`Qwen2.5-14B-Instruct`) using Low-Rank Adaptation (LoRA) [49]. The fine-tuning design was adapted from, and inspired by, the ontology-guided fine-tuning approach reported by Ghazouani et al. (Figure 2.1) [3, 24].

4.5.7.1 Synthetic Training Data Generation

Since no manually annotated corpus of ontology-compliant triples existed for this domain, a synthetic training dataset was generated using the base Qwen2.5-14B-Instruct model in a self-instruction loop. The dataset-generation procedure produced 180 candidate examples in two components: 130 general single-pass examples and 25 consistency pairs. Each consistency pair contains two examples: a text-sourced extraction example and a paraphrased image-caption-style example referring to the same underlying entity. Thus, the 25 pairs contribute 50 examples, yielding $130 + 2 \times 25 = 180$ generated examples.

For each general example, the model first generated a short fictitious French river-restoration scenario and then extracted ontology-constrained subject–predicate–object triples from that scenario. For each consistency pair, the model first extracted triples from a generated text scenario, then generated a short paraphrased caption-style statement referring to the same entity under a different surface form. The canonical entity name from the text example was injected into the entity registry of the caption-style extraction prompt, and the subject label of the resulting caption-style triples was normalized to that canonical name. This pairing was designed to expose the model to the text-to-image coreference pattern used during inference.

4.5.7.2 Dataset Validation and Cleaning

The 180 generated examples were passed through an automated validation and cleaning procedure before fine-tuning. Each triple was retained only if it contained non-empty `subject`, `predicate`, and `object` fields; used a predicate present in the TetraOnto whitelist, and did not use a raw ontology class or predicate label as an entity placeholder. The resulting cleaning procedure therefore enforced JSON structure and predicate-

vocabulary compliance, but did not assess whether individual triples were factually supported by their synthetic source scenarios.

Of the 180 generated examples, 137 were retained and 43 were discarded because they contained malformed JSON or fewer than 50% valid triples after filtering. Across all examples, 2,420 candidate triples were processed; 753 were removed directly because of invalid predicates or placeholder content. The retained corpus contained 1,458 triples, with no remaining out-of-ontology predicates. After direct triple-level filtering, 1,667 triples remained. A subsequent example-level quality filter discarded complete examples containing fewer than 50% valid triples, including 209 otherwise retained triples within those rejected examples, yielding the final corpus of 1,458 triples. The retained examples were serialized in chat-template format for the subsequent 90/10 train–validation split.

4.5.7.3 Training Configuration

The cleaned dataset was converted to a chat-template format pairing a prompt (system and user turns, rendered via the tokenizer's chat template) with a completion (the assistant's JSON triple array, with an explicit end-of-sequence token appended). Fine-tuning used LoRA rank 16, alpha 32, dropout 0.05, applied to all attention and MLP projection matrices (`q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`), on top of the base model loaded in 4-bit NF4 quantization[50] and prepared for k-bit training with gradient checkpointing enabled. Training used a completion-only loss objective (masking the prompt tokens from the loss computation), a maximum sequence length of 8,192 tokens, a learning rate of 2×10^{-4} with linear warmup, gradient accumulation of 2, and a single epoch over a 90/10 train/validation split. The resulting adapter was subsequently merged into the base model weights to produce a single deployable checkpoint, avoiding any adapter-loading overhead at inference time.

4.5.7.4 Training and Inference Schema Scope

The synthetic fine-tuning targets use a reduced three-key representation (`subject`, `predicate`, `object`), whereas the production extraction prompt additionally requests `subject_type`. Fine-tuning therefore directly supervises structured relation extraction and predicate-vocabulary compliance, but does not directly supervise ontology-class prediction. During baseline and fine-tuned inference, the generated `subject_type` is checked against the ontology class whitelist and, when necessary, resolved through the `skos:altLabel` fallback mapping. Consequently, class assignments in the final RDF graphs reflect inference-time model output and deterministic schema validation rather than direct class-label supervision in the synthetic fine-tuning corpus.

4.5.8 Baseline and Fine-Tuned Ablation Protocol

To isolate the effect of the LoRA fine-tuning from confounding pipeline variables, the baseline and fine-tuned extraction runs were executed using a single, byte-identical extraction and RDF-serialization script, differing only in the value of the model checkpoint path (base Qwen2.5-14B-Instruct versus the merged fine-tuned checkpoint) and the output directory. All prompt construction, entity-registry logic, batching behavior, predicate

whitelisting, class resolution, artifact filtering, and entity-merging logic described in Sections 4.5.4–4.5.6 are therefore held constant across both conditions, so that any measured difference in graph quality is attributable solely to the model weights.

4.5.9 Evaluation Metrics

Pipeline 4 is evaluated using Schema Conformance Rate (SCR), Average Node Degree Shift ($\Delta\bar{d}$), and Cross-Modal Link Density (CMLD). An Intrinsic Triple Precision (P_T) audit protocol was defined for factual-grounding assessment but was not completed within this thesis. Metric definitions, computation procedures, and rationale are provided in Section 5.4.6.

Proposed factual-grounding audit. Intrinsic Triple Precision (P_T) is a manual-review protocol estimating the proportion of sampled ontology-conformant triples factually supported by their source passage or verified image caption. It is not reported as a quantitative result because the full annotation pass was not completed.

4.5.10 Pipeline 4 Procedure

The procedural sequence for ontology schema conditioning, generative extraction, programmatic serialization, and cross-modal graph merging is formalized below.

Pseudocode – Pipeline 4: Cross-Modal Knowledge Graph Construction

1. **Input:** Clean text blocks, verified image captions from Pipeline 3, and the master ontology file (`TetraOnto_compact.txt`).
2. Compress the master schema into a minimized structural layout template, isolating authorized classes and property bounds; parse the same file into predicate and class whitelists for downstream validation.
3. Initialize an empty entity registry for the document.
4. **For each source track** $\mathcal{S}$ in {Text Blocks, Image Captions} (image track inherits the registry populated by the text track):
   (a) Split any content unit exceeding the chunk-size threshold into overlapping chunks.
   (b) **For each batch of content units:**
      i. Prepend the compressed schema and the current entity registry to each prompt.
      ii. Prompt the language model to extract relational facts as a flat JSON array of objects containing `subject`, `subject_type`, `predicate`, and `object` fields.
      iii. Parse the response; on JSON failure, attempt bounded salvage repair, else log and skip the block.
      iv. **For each structured triple** (s, t, p, o):

A. Sanitize special characters and apply Unicode normalization.
B. Resolve s's URI via cleaned-label normalization; validate the generated type τ against the ontology class whitelist and, when necessary, attempt resolution through the `skos:altLabel` synonym mapping.
C. Check p against the predicate whitelist; if invalid (or a page-number misclassified as a date), demote the fact to a `skos:note` annotation and continue.
D. Classify o as an entity reference or a literal; assign the predicate as `owl:ObjectProperty` or `owl:DatatypeProperty` accordingly.
E. Annotate generated entity nodes with `:dataSource`, `:extractedFrom`, and `:hasPageNumber` metadata.
F. Append the schema-validated triple to the active `rdflib` graph stream $\mathcal{G}_{\mathcal{S}}$; update the entity registry with any newly seen labels.

5. Serialize the two independent tracks to disk (`texts.ttl`, `images.ttl`).

6. Merge the two graphs into a single file; apply the three-stage entity resolution process (exact-label merge, LLM-mediated alias resolution, similarity-based alias merge) to fuse cross-modal duplicates.

7. Serialize the resolved merged graph (`merged_knowledge_graph.ttl`) and compute the corpus-level evaluation metrics defined in Section 4.5.9.

4.5.11 Summary and Experimental Evaluation

This chapter described the end-to-end multi-modal knowledge graph extraction framework and its four pipelines. Pipeline 1 performs document layout parsing and structural noise filtration. Pipeline 2 implements late-interaction cross-modal retrieval to re-anchor figures to localized narrative context. Pipeline 3 uses two-phase visual captioning to identify image-derived physical indicators while reducing language-prior hallucinations. Finally, Pipeline 4 formalizes programmatic serialization and cross-modal reconciliation under the `TetraOnto` domain schema.

Chapter 5 presents the empirical evaluation of the complete system, including the French Rhine restoration corpus, retrieval, captioning, and graph-construction metrics, and ablation studies validating each design choice.

Chapter 5

Experiments and Evaluation

5.1 Objectives of the Experimental Study

The experimental study pursues four objectives, each corresponding to one pipeline:

1. **Pipeline 1 - Document Extraction**: Evaluate whether layout-aware, rasterization-based image extraction produces more coherent figure crops compared to the native XObject-based approach, and measure the noise rejection rate of the deterministic multi-stage structural filtration framework.

2. **Pipeline 2 - Image–Text Matching**: Conduct a full ablation study across six retrieval models from two architectural families (single-vector and multi-vector late interaction), evaluating their ability to retrieve the correct grounding text block for each extracted image.

3. **Pipeline 3 - Image Captioning**: Conduct a six-stage ablation study across (a) seven candidate VLMs under no-prompt and structured-prompt conditions, (b) three context-injection strategies (unblended structured, parallel, and blended fusion), (c) six advanced prompt-engineering strategies applied to the best-performing configuration, and (d) four contrastive decoding configurations, measuring both caption fidelity and caption novelty.

4. **Pipeline 4 - Knowledge Graph Construction**: Evaluate the enrichment contribution of image-derived triples over text-only triples, measuring entity connectivity gain and graph expansion.

5.2 Dataset Description

All experiments used a corpus of five heterogeneous French technical and institutional documents covering Rhine catchment restoration, river hydromorphology, ecological conservation, thermal habitat assessment, and historical river evolution. The corpus was provided by the project partner. For consistent traceability throughout this thesis, the documents are referred to exclusively by the canonical identifiers `PDF-1` to `PDF-5`, defined below.

The source PDFs are archived in the project repository[1]. PDFs 1, 3, 4, and 5 concern the Rhine basin restoration contexts and PDF 2 was included for its structurally diverse French hydro-ecological content rather than for geographical relevance to the Rhine basin.

[1] Project repository

- **PDF-1:** ***Étude hydromorphologique du ruisseau de Norroy.*** A hydromorphological study of the Norroy stream. A fixed evaluation subset of 25 non-decorative, information-bearing figures was selected after Pipeline 1 extraction and structural-noise filtering.
- **PDF-2:** ***Fiche technique: Définition, identification, préservation et restauration des zones de sources des cours d'eau en Bretagne et Pays de la Loire.*** A technical guidance document addressing the definition, identification, preservation, and restoration of headwater-source zones in Brittany and Pays de la Loire. The fixed evaluation set contains 18 image assets.
- **PDF-3:** ***Dossier de protection des espèces de la Zorn.*** An administrative and ecological species-protection dossier supporting a 2021–2025 river-restoration programme led by the Communauté d'Agglomération de Haguenau. The programme includes ecological restoration works, fauna–flora diagnosis, and protected-species impact assessment. The fixed evaluation set contains 19 image assets.
- **PDF-4:** ***Contribution à la caractérisation du potentiel du Vieux Rhin pour les grands salmonidés migrateurs: métabolisme thermique, cartographie des habitats et données piscicoles disponibles.*** A Master 2 internship report in natural-environment engineering by Floriane Kuntzler. It examines the potential of the Vieux Rhin for large migratory salmonids through thermal metabolism, habitat mapping, and available fish-data analysis. The fixed evaluation set contains 38 image assets.
- **PDF-5:** ***Carte à la une: reconstituer le Rhin disparu.*** A historical-geography article for educators that reconstructs two centuries of Rhine river transformation through archival cartography. The fixed evaluation set contains 5 image assets.

Table 5.1: Authoritative composition of the five-document evaluation corpus. Image counts in the final column define the fixed image set used for retrieval, captioning, and image-derived knowledge-graph extraction throughout Chapter 5.

ID	Short label	Pages	Final evaluation images
PDF-1	Norroy study (Fichier)	23	25
PDF-2	OFB Fiche Technique	41	18
PDF-3	Zorn species dossier	35	19
PDF-4	Vieux Rhin report (Rapport Floriane)	55	38
PDF-5	Rhin Disparu	8	5
Total		162	**105**

All subsequent references, table headers, figure captions, and result discussions use the identifiers `PDF-1` through `PDF-5`. The corpus contains 105 images in total for Pipeline 3 captioning evaluation.

Ground-truth image–text associations were manually annotated for Pipeline 2 evaluation. All 105 captioning images across the five documents were used for Pipeline 3

evaluation. VQA Consistency and Atomic Score metrics were computed using an automated VLM-based atomic claim-verification protocol, rather than manually authored question–answer pairs.

Document acquisition was constrained by institutional data availability; despite efforts to source a thematically diverse set of restoration interventions (e.g., fish-pass construction, sediment management), the final corpus is concentrated around a smaller number of well-documented sites and intervention types. This affects the generalizability of extraction and enrichment results, which should be read as intervention-type-specific findings rather than domain-wide performance guarantees.

5.3 Evaluation Metrics for Text Retrieval

Let Q denote the set of evaluated image queries. For each query $q \in Q$, let G_q be the set of manually annotated relevant text blocks and let $R_q^{(5)}$ denote the ranked set of up to five text blocks returned by a retrieval model. Relevance is binary: a retrieved block is relevant if and only if it belongs to G_q. The following metrics are computed independently for each query and then averaged within each document.

5.3.1 Recall@5

Because a query may have multiple relevant text blocks, Recall@5 measures the proportion of its annotated relevant blocks recovered among the top five retrieved results:

$$\text{Recall@5}(q) = \frac{\left|R_q^{(5)} \cap G_q\right|}{|G_q|}.$$

This fractional formulation assigns partial credit when a model retrieves some, but not all, of the relevant blocks.

5.3.2 Precision@1

Precision@1 measures whether the top-ranked retrieved block is relevant:

$$\text{P@1}(q) = \begin{cases} 1, & \text{if the rank-1 block belongs to } G_q, \\ 0, & \text{otherwise.} \end{cases}$$

5.3.3 Mean Reciprocal Rank (MRR@5)

Let r_q^* be the rank of the first relevant block returned for query q. The reciprocal-rank contribution is:

$$\text{RR@5}(q) = \begin{cases} \frac{1}{r_q^*}, & \text{if a relevant block is retrieved in ranks 1–5,} \\ 0, & \text{otherwise.} \end{cases}$$

Mean Reciprocal Rank@5 is the average of RR@5(q) over all evaluated queries.

5.3.4 Normalized Discounted Cumulative Gain (nDCG@5)

nDCG@5 evaluates the rank ordering of all relevant blocks within the top five retrieved results. Let $\mathrm{rel}_q(r) = 1$ when the block at rank r belongs to G_q, and 0 otherwise. Then:

$$\mathrm{DCG@5}(q) = \sum_{r=1}^{5} \frac{\mathrm{rel}_q(r)}{\log_2(r+1)}.$$

The ideal discounted cumulative gain is:

$$\mathrm{IDCG@5}(q) = \sum_{r=1}^{\min(|G_q|,5)} \frac{1}{\log_2(r+1)}.$$

The normalized score is:

$$\mathrm{nDCG@5}(q) = \frac{\mathrm{DCG@5}(q)}{\mathrm{IDCG@5}(q)}.$$

Among the retrieval metrics reported in this work, macro-nDCG@5 serves as the primary model-selection criterion because it jointly evaluates the recovery and ranking of multiple annotated relevant blocks. Recall@5, Precision@1, and MRR@5 are reported as complementary diagnostic measures.

5.4 Evaluation Metrics for Image Captioning

Standard n-gram metrics (BLEU, ROUGE) are insufficient for evaluating VLM-generated technical captions, as they penalize domain-specific elaboration and paraphrase. The following metrics are used.

5.4.1 CLIPScore

Measures semantic alignment between a generated caption $\hat{c}$ and the source image $\mathcal{I}$ in CLIP's joint embedding space [51]:

$$\mathrm{CLIPScore}(\hat{c}, \mathcal{I}) = w \cdot \max(0,\ \cos(f_I(\mathcal{I}),\ f_T(\hat{c})))$$

Higher scores indicate better visual grounding of the caption.

5.4.2 PAC-S

PAC-S (Positive Alignment Caption Score) [52] extends CLIPScore by improving the image–text representation through positive-augmented contrastive learning, then measures alignment between the candidate caption and the source image. PAC-S was similarly discontinued for the same reasons as CLIPScore: as captions grow in factual and structural complexity, reference-free alignment metrics based on vision-language embeddings trained on short web captions struggle to reflect the quality of specialised scientific descriptions.

Note on discontinuation: CLIPScore and PAC-S were used during early pipeline development but were discontinued as primary evaluation metrics. As caption complexity increases, both metrics saturate and lose discriminative power: a highly detailed, domain-specific scientific caption is penalised relative to a short generic description, as the CLIP text encoder was not trained on technical register text [53]. All final ablation comparisons therefore rely exclusively on NLI-based novelty and grounding scores, VQA consistency, and enrichment rate.

5.4.3 NLI-Based Caption Analysis

Grounding is assessed using a SummaC-style zero-shot entailment procedure [54], adapted to compare a generated caption $\hat{c}$ against a retrieved reference context r extracted from the source document. Both the reference context and the caption are segmented into sentences. For each caption sentence, a pretrained natural language inference (NLI) model computes an entailment probability against every reference sentence, and the maximum probability across reference sentences is retained; these per-sentence maxima are then averaged across all caption sentences to yield a single continuous grounding score in $[0, 1]$:

$$\text{Grounding}(\hat{c}, r) = \frac{1}{|\hat{c}|} \sum_{s \in \hat{c}} \max_{t \in r} P_{\text{entail}}(t, s)$$

where $\hat{c}$ and r denote the sets of sentences obtained from the caption and the reference context, respectively. A complementary text-non-entailment score is defined as $1 - \text{Grounding}(\hat{c}, r)$, measuring the extent to which caption content is not directly entailed by the retrieved context. This score alone does not distinguish visually supported image-derived information from unsupported content. It is therefore interpreted jointly with the Phase 1 VQA atomic score and the Verified Enrichment Rate defined in Section 5.4.4.1. For interpretability in reporting, the continuous grounding score is additionally bucketed into three descriptive categories, entailment ($\geqslant 0.7$), neutral ($\geqslant 0.4$), and contradiction (< 0.4), though these thresholds are used only for summary reporting and do not affect any quantitative scoring computation.

5.4.4 VQA-Consistency Score

Caption faithfulness is assessed through an automated VQA-style atomic claim decomposition and direct visual verification procedure, rather than a paired human-annotated question–answer comparison. Each structured caption is first split into its Phase 1 (purely visual observation) and Phase 2 (contextual enrichment) segments. A vision-language judge model (Qwen2-VL-7B) decomposes each phase into a small set of atomic, independently verifiable claims: 3–5 claims for Phase 1 visual content and 2–4 claims for Phase 2 contextual content. Each atomic claim is then presented to the same judge model together with the source image and, for Phase 2 claims, a short excerpt of retrieved document context. The model is instructed to describe the relevant evidence and issue a binary verdict: `CONFIRMÉ` or `NON-CONFIRMÉ`.

This produces a $\mathrm{VQA}_{\mathrm{P1}}$ score for every caption and, where Phase 2 is present, a $\mathrm{VQA}_{\mathrm{P2}}$ score. When both phases are available, the combined score is their arithmetic mean:

$$\mathrm{VQA}_{\mathrm{combined}} = \frac{\mathrm{VQA}_{\mathrm{P1}} + \mathrm{VQA}_{\mathrm{P2}}}{2}.$$

The metric is an automated, model-based proxy for caption grounding, rather than accuracy on a human-annotated VQA benchmark.

5.4.4.1 Verified Enrichment Rate

The Verified Enrichment Rate estimates the proportion of image captions that introduce content not directly entailed by the retrieved textual context while retaining a minimum level of visual support. For each image i, the NLI grounding score $G_i \in [0, 1]$ is computed between its generated caption and retrieved context. Captions are categorized as neutral when $0.4 \leqslant G_i < 0.7$, indicating that their content is neither strongly entailed by nor strongly contradicted by the retrieved text. Independently, the Phase 1 VQA atomic score V_i measures the proportion of decomposed directly observable claims confirmed against the image pixels.

An image is counted as a verified enrichment when its caption is NLI-neutral and its Phase 1 VQA score reaches at least 0.5:

$$\mathrm{VE}_i = \mathbb{1}\left[0.4 \leqslant G_i < 0.7 \ \wedge \ V_i \geqslant 0.5\right].$$

For an evaluation set of N images, the Verified Enrichment Rate is computed as:

$$\mathrm{EnrichmentRate} = \frac{1}{N} \sum_{i=1}^{N} \mathrm{VE}_i.$$

This operational definition treats enrichment as caption content that is not directly recoverable from the retrieved text while meeting a minimum image-grounding threshold. It remains an automated proxy rather than a human factual-accuracy audit; its interpretation should therefore be considered alongside the underlying NLI and VQA scores.

5.4.5 Weighted Multi-Criteria Model Selection Score

Model selection across ablations is performed using a weighted multi-criteria scoring framework. This approach is motivated by the fact that quality, inference speed, and model size represent competing deployment objectives with no universally preferred trade-off. A weighted linear scalarisation combines the normalised criteria into a single composite score:

$$\mathrm{Score} = w_1 \cdot Q_{\mathrm{norm}} + w_2 \cdot S_{\mathrm{norm}} + w_3 \cdot Z_{\mathrm{norm}}$$

where Q_{norm} is the normalised quality score, built from VQA Consistency, Enrichment Rate, and NLI Text-Non-Entailment; S_{norm} is the normalised observed end-to-end throughput score, obtained by inverting seconds per image; and Z_{norm} is the normalised model compactness score, obtained by inverting parameter count. Within each ablation, each criterion is min–max normalised to the interval $[0, 1]$ using bounds computed over

the candidate configurations evaluated in that ablation. The resulting composite score is used only to rank candidates within the corresponding ablation and is not interpreted as directly comparable across separate ablation rounds. Any candidate for which throughput or parameter count cannot be reliably determined is excluded from the ranking rather than assigned a default value, since an unresolved measurement defaulted to an artificially favourable value would distort its normalised score and compromise the validity of the comparison.

The following weighting scheme is used consistently across all ablation rounds. It prioritises verified extraction quality and information discovery while retaining observed throughput and model compactness as deployment-relevant operational constraints:

- Quality sub-weights: VQA Consistency = 0.50, Enrichment Rate = 0.35, NLI Text-Non-Entailment = 0.15;
- Outer weights: Quality = 0.60, Speed = 0.25, Size = 0.15.

These weights express the deployment priorities of this study rather than universal model preferences; the underlying raw metrics are reported separately to make alternative trade-offs transparent. For final deployment selection, all executed candidate configurations are pooled into a separate global comparison set. Quality, observed throughput, and model size are re-normalised over this full set before computing a final cross-ablation composite score. Only this final global ranking is used to compare configurations originating from different ablation rounds.

Weighted linear scalarisation is a standard approach for aggregating multiple normalised objectives according to explicitly selected decision weights [55].

Inference Timing Protocol. Latency is reported as observed end-to-end inference throughput in seconds per image for each evaluated configuration. Each experimental run was executed on a dedicated allocated compute node with the GPU resources required by the corresponding model; measured jobs did not share an active GPU with concurrent workloads. Timing values were verified through two or three repeated executions of the same configuration. Larger models were executed under 4-bit quantisation where required by VRAM constraints, whereas smaller models were run without this quantisation; numerical precision is therefore treated as part of the deployable configuration rather than as a nuisance variable. Consequently, reported latency reflects the operational cost of the complete evaluated configuration, including model choice, allocated hardware resources, numerical precision, prompt and context format, image processing, and generated output length, rather than an architecture-intrinsic speed benchmark. Hardware allocation, quantisation configuration, and execution details for each run are recorded in the accompanying implementation repository.

5.4.6 Knowledge Graph Integration Metrics

To quantitatively assess vocabulary alignment, structural enrichment, and multi-modal connectivity, three metrics are computed independently over the text-only graph, the image-only graph, and the final merged graph produced for each document. A fourth

measure, Intrinsic Triple Precision, is specified as a pending manual factual-grounding audit protocol and is not reported as a computed result in this thesis.

5.4.6.1 Schema Conformance Rate (SCR)

The Schema Conformance Rate measures the proportion of domain-relation facts in the merged graph whose predicate belongs to the authorized ontology vocabulary, as opposed to having been demoted to an annotation during serialization (Section 4.5.5). Let $\mathcal{T}_{\text{merged}}$ denote the full set of triples in the merged graph, and let $\mathcal{P}_{\text{prov}} = \{\texttt{:dataSource}, \texttt{:hasPageNumber}, \texttt{:extractedFrom}\}$ denote the fixed set of provenance and bookkeeping predicates that the pipeline attaches automatically to every individual and that are therefore excluded from the conformance calculation, since their presence is guaranteed by construction rather than reflective of the model's own predicate choices. We define the conformant count C as the number of triples whose predicate lies in the domain ontology namespace and outside $\mathcal{P}_{\text{prov}}$, and the demoted count D as the number of triples whose predicate is `skos:note` (the annotation slot to which non-conformant, model-generated predicates are redirected rather than discarded):

$$C = \left|\{(s,p,o) \in \mathcal{T}_{\text{merged}} : p \in \mathcal{O}_P,\ p \notin \mathcal{P}_{\text{prov}}\}\right|,$$
$$D = \left|\{(s,p,o) \in \mathcal{T}_{\text{merged}} : p = \texttt{skos:note}\}\right|.$$

$$\text{SCR} = \frac{C}{C + D}$$

where $\mathcal{O}_P$ denotes the set of predicate URIs declared under the domain ontology namespace. SCR therefore measures conformance at the level of individual triple instances rather than unique predicate types, so a document containing many repeated uses of a single non-conformant predicate is penalized proportionally to its actual frequency of occurrence in the extracted output, not merely once. Because $\mathcal{P}_{\text{prov}}$ is excluded from both C and D, SCR isolates exactly the subset of predicate decisions attributable to the generative model, rather than being diluted by the fixed, deterministically-added structural metadata every individual receives regardless of extraction quality.

5.4.6.2 Average Node Degree Shift ($\Delta\bar{d}$)

The Average Node Degree Shift quantifies the change in URI-level graph density between the text-only graph and the final merged graph. For a graph G, let $V(G)$ denote the set of unique URI-typed nodes appearing as a subject or as a URI-valued object in G, excluding literal-valued nodes, and let $E_{\text{URI}}(G)$ denote the set of triples whose object is a URI resource rather than a literal. The average node degree is defined as:

$$\bar{d}(G) = \frac{2 \cdot |E_{\text{URI}}(G)|}{|V(G)|}$$

The degree shift between the merged and text-only graphs is:

$$\Delta\bar{d} = \bar{d}(\mathcal{G}_{\text{merged}}) - \bar{d}(\mathcal{G}_{\text{text}})$$

Because $E_{\text{URI}}(G)$ includes both inter-entity relations and URI-valued typing assertions, this measure reflects overall URI-level structural densification rather than validated semantic connectivity alone. A positive $\Delta\bar{d}$ is therefore consistent with increased URI-level graph density following the incorporation of image-derived content.

5.4.6.3 Cross-Modal Link Density (CMLD)

The Cross-Modal Link Density quantifies the extent to which text-derived and image-derived information is genuinely fused at the graph level, rather than coexisting as disconnected, single-modality fragments. Let $\mathcal{E}_{\text{text}}$ and $\mathcal{E}_{\text{image}}$ denote the sets of `owl:NamedIndividual` entities declared in the standalone text graph and standalone image graph, respectively, prior to merging. The overlap set is defined as their intersection:

$$\mathcal{E}_{\text{overlap}} = \mathcal{E}_{\text{text}} \cap \mathcal{E}_{\text{image}}$$

This intersection is produced by the pipeline's cross-modal coreference and entity-resolution mechanisms. Entity URIs are first derived deterministically from normalized surface labels (Section 4.5.6), and the two-pass extraction procedure shares a running entity registry across the text and image passes (Section 4.5.4). These mechanisms allow entities mentioned in both modalities under the same canonical label to receive matching URIs before graph merging. After the standalone graphs are merged, the three-stage entity-resolution procedure described in Section 4.5.6 additionally resolves exact-label, alias, and near-duplicate matches. The overlap set E_{overlap} therefore measures the cross-modal overlap realized by this complete coreference and entity-resolution procedure.

$$\text{CMLD} = \frac{\left|\{(s,p,o) \in \mathcal{T}_{\text{merged}} : p \in \mathcal{O}_P \backslash \mathcal{P}_{\text{prov}},\ (s \in \mathcal{E}_{\text{overlap}} \vee o \in \mathcal{E}_{\text{overlap}})\}\right|}{C}$$

where C is the same conformant-triple count used in the SCR computation. Normalizing by C rather than by the total triple count $|\mathcal{T}_{\text{merged}}|$ ensures that CMLD measures cross-modal fusion specifically among substantive domain-relation edges, without being diluted by the fixed volume of provenance and labeling triples that every individual receives irrespective of its modality.

5.4.6.4 Intrinsic Triple Precision (PT)

Since SCR and CMLD evaluate structural and vocabulary-level properties of the graph, neither is capable of detecting a triple that uses a valid, whitelisted predicate but asserts a fact not actually supported by its source text or caption. To audit this class of error, a fixed-seed random sample (without replacement, up to 50 triples) is drawn from the conformant domain-relation triples of the merged graph for each document and pipeline condition, using the same predicate-namespace filter applied in the SCR and CMLD computations to exclude provenance and bookkeeping edges from the sampling pool. Each sampled triple is exported in a human-readable form (shortened subject, predicate, and object labels) for manual annotation, in which a reviewer marks the triple as True (factually supported by the source passage), False (unsupported or fabricated), or Unsure. The Intrinsic Triple Precision is then computed as:

$$\text{PT} = \frac{n_{\text{true}}}{n_{\text{sampled}}}$$

Because this metric requires human judgment against the original source material, it is reported as a pending manual-review protocol in this work (Section 5.8.2.2) rather than as a fully executed measurement across the entire evaluation corpus.

5.5 Pipeline 1: Extraction and Layout Filtering Results

5.5.1 Multi-Stage Structural Noise Filtering

The multi-stage structural filtration framework detailed in Section 4.2, combining cross-page n-gram profiling, text-normalization fuzzy matching, and deep vision-based block classification via DocLayout-YOLOv10, was evaluated across the target environmental report corpus to measure its efficacy against structural document noise.

The boilerplate identification procedure, based on n-gram profiling and Gestalt-based fuzzy matching (Section 4.2), was applied uniformly across the corpus to isolate repetitive institutional noise such as headers, footers, and running page banners.

Quantitative filtration metrics across the primary evaluation datasets are documented in Table 5.2.

Table 5.2: Multi-stage structural and layout filtering metrics across the datasets defined in Section 5.2.

Document Source	Total Pages Filtered	TOC Entries Repeaters	Boilerplate Removed	YOLO Abandon Regions	Finalized Blocks Retained
Document 1 (*PDF-1*)	23	8	42	15	74
Document 2 (*PDF-2*)	41	6	28	11	51
Document 3 (*PDF-3*)	35	4	12	7	38
Document 4 (*PDF-4*)	55	18	84	29	142
Document 5 (*PDF-5*)	8	2	9	4	18

The programmatic evaluation reveals that structural layout noise accounts for nearly half (approximately 46%) of all raw document segments across the corpus, ranging from 38% to 48% depending on document type. Using the DocLayout-YOLOv10 configuration described in Section 4.2, the pipeline successfully identifies uninformative graphic zones categorized as `abandon` regions. Filtering these elements prior to downstream processing reduces narrative pollution, helping ensure that primarily high-value, domain-specific text transitions to the knowledge graph construction phase.

5.5.2 Cross-Modal Deduplication and Visual Layout Extraction

A primary bottleneck in multi-modal information extraction is text duplication caused by the simultaneous extraction of figure captions as regular text blocks and as image components. To address this, the pipeline routes extracted visual regions into a token-based cross-modal deduplication framework powered by the DocTR OCR predictor toolkit[35]. The programmatic text-extraction backend applied here follows exactly the configuration detailed in Section 4.2.

A qualitative validation was conducted on Document 3 to compare this layout-aware approach against traditional object-stream extraction models. On Page 2, a complex, four-panel historical river evolution map spanning the years 1828 to 2010, the legacy XObject

parser failed to recognize the layout structure. It extracted 16 separate, fragmented image pieces from the singular graphic panel, separating the visual data from its associated legend.

In contrast, the DocLayout-YOLO framework successfully mapped the entire asset as a unified coordinate entity. By expanding boundaries, merging the panel gaps, and pulling in the lower caption segment ("Figure 2: Évolution du paysage fluvial dans le secteur du Rohrschollen entre 1828 et 2010"), the model produced a single, high-fidelity visual asset. The subsequent DocTR pass verified the embedded tokens, successfully deduplicating the caption from the primary narrative stream and supporting the conclusion that layout-aware vision-language parsing is required for technical environmental dossiers.

Having established that the extraction stage reliably isolates clean text blocks and unified figure assets, the following section evaluates how effectively these two modalities can be reconnected through cross-modal retrieval.

5.6 Pipeline 2: Image–Text Matching Results

This section presents the empirical evaluation of the six candidate retrieval architectures introduced in Section 4.3, applied to the text blocks and isolated images produced by Pipeline 1. The objective is to identify the model configuration that most reliably recovers the correct narrative context for each extracted image, which subsequently drives the grounded captioning stage described in Pipeline 3.

5.6.1 Experimental Setup

Six cross-modal retrieval models were evaluated across five documents(see Section: 5.2) to represent distinct document layouts and content types. The compiled benchmarking corpus comprises a collective total of 105 manually annotated ground-truth image-text matching pairs evaluated under zero-caption and zero-shot parameters.

The retrieval quality is tracked using four standard information retrieval metrics(5.3. All multi-vector architectures utilize a mean-normalized MaxSim late-interaction operator to ensure score equity across variable paragraph lengths, whereas all single-vector baselines compute visual-semantic matching alignment via standard cosine similarity matrices.

5.6.2 Retrieval Ablation Results

Table 5.3 displays the complete breakdown of individual retrieval metrics across both architectural families.

5.6.3 Global Leaderboard

To weight each document domain equally regardless of image count, a global macro-averaging protocol was implemented. Table 5.4 outlines the unified macro-averages calculated systematically across all five processed files, serving as the central validation leaderboard for model selection.

Table 5.3: Cross-modal text-image retrieval metrics for single-vector and multi-vector model families across five independent evaluation documents. Best result per document in bold.

Document	Family	Model	Recall@5	P@1	MRR	nDCG@5
PDF-1 ($N = 25$)	Single-Vector	CLIP (baseline)	0.1403	0.0800	0.2527	0.1455
		DSE-Qwen2	0.2996	0.5600	0.6180	0.3741
		VIDORE-VDR	**0.3267**	**0.6400**	**0.6800**	**0.4221**
	Multi-Vector	COLPALI	0.2410	0.2000	0.3867	0.2565
		COLQWEN	**0.3748**	0.3200	0.5347	**0.3974**
		NEMOTRON	0.2779	**0.5200**	**0.6167**	0.3438
PDF-2 ($N = 18$)	Single-Vector	CLIP (baseline)	0.1167	0.1765	0.2667	0.1266
		DSE-Qwen2	**0.5794**	0.9412	0.9706	**0.6848**
		VIDORE-VDR	0.5598	**1.0000**	**1.0000**	0.6836
	Multi-Vector	COLPALI	0.4951	0.5294	0.7098	0.5015
		COLQWEN	0.4706	0.6471	0.7990	0.5265
		NEMOTRON	**0.5549**	**0.7647**	**0.8725**	**0.6142**
PDF-3 ($N = 19$)	Single-Vector	CLIP (baseline)	0.1216	0.1053	0.2254	0.1183
		DSE-Qwen2	0.4416	**0.7368**	**0.8395**	0.5289
		VIDORE-VDR	**0.4803**	**0.7368**	0.8289	**0.5689**
	Multi-Vector	COLPALI	0.3044	0.5263	0.6360	0.3735
		COLQWEN	0.3945	0.4737	0.6070	0.4296
		NEMOTRON	**0.5337**	**0.7368**	**0.8684**	**0.6202**
PDF-4 ($N = 38$)	Single-Vector	CLIP (baseline)	0.1826	0.2105	0.3101	0.1747
		DSE-Qwen2	0.3955	0.6316	0.6974	0.4823
		VIDORE-VDR	**0.4945**	**0.6579**	**0.7276**	**0.5445**
	Multi-Vector	COLPALI	0.4034	0.5789	0.6610	0.4594
		COLQWEN	0.4504	0.6053	0.6886	0.5141
		NEMOTRON	**0.4588**	**0.8158**	**0.8737**	**0.5623**
PDF-5 ($N = 5$)	Single-Vector	CLIP (baseline)	0.2250	0.0000	0.3167	0.1848
		DSE-Qwen2	**0.6750**	**0.8000**	**0.9000**	**0.7453**
		VIDORE-VDR	**0.6750**	0.6000	0.8000	0.7097
	Multi-Vector	COLPALI	**0.3250**	**0.2000**	0.4067	0.3487
		COLQWEN	**0.3250**	**0.2000**	**0.5000**	**0.3868**
		NEMOTRON	0.2000	**0.2000**	0.3000	0.2333

Table 5.4: Global Leaderboard: Arithmetic macro-averages calculated across all five document domains, sorted by the primary target ranking quality metric (nDCG@5).

Family	Model	Macro-Recall@5	Macro-P@1	Macro-MRR	Macro-nDCG@5
Single-Vector	VIDORE-VDR	**0.5073**	0.7269	**0.8073**	**0.5858**
	DSE-QWEN2	0.4782	**0.7339**	0.8051	0.5631
Multi-Vector	NEMOTRON	**0.4051**	**0.6075**	**0.7063**	**0.4748**
	COLQWEN	0.4031	0.4492	0.6259	0.4509
	COLPALI	0.3538	0.4069	0.5600	0.3879
Baseline	CLIP	0.1572	0.1145	0.2743	0.1500

5.6.4 Interpretation and Retriever Selection

The CLIP baseline establishes a conservative lower bound for this task. Its weak performance should not be attributed solely to single-vector pooling: candidate text blocks are truncated to CLIP's 77-token limit and, unlike the other models, do not receive hybrid page-proximity re-weighting ($\beta = 0.02$). The result therefore shows that this general-purpose, short-context CLIP configuration is not suitable for retrieving paragraph-length narrative context from the extracted figures; it does not isolate the separate effects of pooling, truncation, positional re-weighting, and document-specific pretraining.

The comparison within the multi-vector family highlights that retrieving one relevant block and recovering the broader set of relevant context are different retrieval behaviours. COLPALI achieves lower Recall@5 than COLQWEN, indicating less consistent recovery of all annotated relevant blocks within the top-five window. NEMOTRON ColEmbed provides the strongest overall late-interaction performance, but the results also show that late interaction alone is not sufficient to determine the best retriever for this corpus.

The main finding is that document-specialised training has a larger practical effect than the architectural distinction between single-vector and multi-vector retrieval in this evaluation. Although late interaction was initially expected to better preserve fine-grained document evidence, VIDORE-VDR achieves the strongest macro-level ranking performance. Its advantage on nDCG@5 is particularly important because Pipeline 3 requires relevant context to appear early in the ranked list, rather than merely somewhere among the retrieved candidates. DSE-QWEN2 remains competitive and obtains the strongest rank-one result, but VIDORE-VDR provides the more consistent top-five context ranking used by the downstream captioning stage.

Accordingly, `vdr-2b-multi-v1` is selected for Pipeline 3. This selection is specific to the present corpus and evaluation protocol, and does not establish that single-vector retrieval is generally superior to late-interaction retrieval. All results should also be interpreted in light of the internally annotated ground-truth associations, which were not independently adjudicated by domain experts.

5.7 Pipeline 3: Image Captioning Results

This section presents the results of the six-stage ablation study conducted to optimise the image captioning pipeline. The ablations progress from a no-prompt baseline through increasingly structured prompting and context strategies, before narrowing to advanced prompt engineering and decoding-level interventions applied to the single best-performing configuration, culminating in the global cross-ablation synthesis that determines the final production pipeline.

> *Note*: Retrieval metrics are macro-averaged by document to prevent large PDFs from dominating retriever selection, whereas captioning and enrichment metrics are pooled across all evaluated images because the individual image-caption pair is the evaluation unit.

5.7.1 Ablation 1: No-Prompt Baseline

Ablation 1 establishes the lower-bound baseline for the full captioning pipeline by evaluating all seven candidate vision-language models (VLMs) without a structured prompt or retrieval context. Each model receives only the raw image and a minimal instruction ("Décris cette image."). This configuration isolates the intrinsic captioning capability of each backbone in the absence of the two-phase prompting strategy, contextual grounding, and document-level summarization introduced in subsequent ablations.

Because Ablation 1 injects no retrieval context into the generation prompt, its NLI grounding and enrichment scores are nevertheless evaluated against the same retrieval-matched context blocks used in later ablations (Section 4.3). This keeps the evaluation reference constant while varying only the model input, enabling estimation of the contribution of structured prompting and retrieval context.

Unlike Ablations 2 through 4, this baseline is not part of the model-selection funnel. Model selection proceeds exclusively from Ablation 2 through Ablation 4, which progressively narrow the candidate pool to the single best-performing configuration carried forward into the prompt-engineering and decoding-level ablations. Ablation 1 is therefore a diagnostic control used to quantify the contribution of structured prompting and retrieval context, rather than to select a production model.

5.7.1.1 Baseline Captioning Metrics

Table 5.5 presents the raw evaluation metrics for all seven models under the no-prompt baseline configuration. The model order matches the ranking established in Ablation 2 to facilitate direct cross-reference.

Table 5.5: Ablation 1 raw evaluation metrics: comparison of seven VLMs under the no-prompt baseline.

Model	Published Params (B)	s/img	NLI Novelty	Comb VQA	Enrich
Mistral-Small-3.1-24B	24.0B	81.9	0.892	0.885	0.038
NVILA-Lite-8B	8.1B	14.7	0.911	0.857	0.019
Qwen3-VL-30B	30.3B	51.7	0.916	0.910	0.019
Pixtral-12B	12.4B	75.2	0.870	0.894	0.067
Llama-3.2-Vision-11B	11.0B	38.3	0.942	0.841	0.019
Qwen3.6-27B	27.0B	4.9	0.894	0.825	0.000
InternVL2.5-8B	8.1B	30.0	0.860	0.877	0.095

5.7.1.2 Findings and Analysis

The no-prompt baseline establishes each model's captioning behaviour when it has access only to the source image. In this setting, captions are typically short and generic, with no document retrieval context to support contextual enrichment. The baseline results are compared directly with the structured-prompt condition in Ablation 2, where the paired

analysis isolates the effect of introducing the two-phase prompt and VidoRe retrieval context.

Although Ablation 1 is not used for model selection, it provides the necessary lower-bound control for interpreting the later results. In particular, it allows the observed changes in enrichment, NLI novelty, and VQA consistency under the structured configuration to be attributed to prompt and context injection rather than to inherent differences between the candidate architectures.

5.7.2 Ablation 2: Structured Prompting and Retrieval Context

Ablation 2 evaluates all seven candidate vision-language models (VLMs) under the two-phase structured prompt described in Section 4.4.3, with VidoRe retrieval context (Section 5.6) across five documents.

5.7.2.1 Evaluation Metrics

Table 5.6 presents the raw quality, throughput, and published-parameter values for all seven candidate models under the two-phase structured prompt with VidoRe context.

Table 5.6: Ablation 2 raw evaluation metrics: cross-comparison of seven VLMs under a two-phase structured prompt with VidoRe context.

Model	Published Params (B)	s/img	NLI Novelty	Comb VQA	Enrichment
Mistral-Small-3.1-24B	24.0B	61.2	0.808	0.841	0.238
NVILA-Lite-8B	8.1B	14.3	0.846	0.795	0.095
Qwen3-VL-30B	30.3B	51.8	0.879	0.927	0.038
Pixtral-12B	12.4B	41.0	0.805	0.783	0.086
Llama-3.2-Vision-11B	11.0B	82.7	0.837	0.802	0.086
Qwen3.6-27B	27.0B	107.0	0.925	0.854	0.000
InternVL2.5-8B	8.1B	20.6	0.823	0.546	0.067

5.7.2.2 Effect of Structured Prompting and Retrieval Context

Table 5.7 compares the Ablation 2 configuration with the no-prompt baseline in Ablation 1. The same seven models are evaluated with the same evaluation pipeline and retrieval-matched reference context; the generation condition differs only through the addition of the two-phase structured prompt and VidoRe retrieval context in Ablation 2. The paired comparison therefore controls for backbone architecture and isolates the effect of structured prompting and retrieval-context injection.

The paired comparison shows that structured prompting combined with retrieved context changes captioning behaviour consistently across most candidate models. Verified enrichment increases on five of the seven models, raising the mean enrichment rate from 0.037 under the no-prompt baseline to 0.087 under Ablation 2, a relative increase of approximately 137%. The largest gain is observed for `Mistral-Small-3.1-24B`, whose enrichment rate increases by 0.200. This is the central result of the control comparison:

Table 5.7: Paired comparison of raw quality metrics between Ablation 1 and Ablation 2, per model. Δ columns reports the difference.

	NLI (Novelty)			VQA			Enrichment		
Model	**Abl.1**	**Abl.2**	**Δ**	**Abl.1**	**Abl.2**	**Δ**	**Abl.1**	**Abl.2**	**Δ**
Mistral-Small-3.1-24B	0.892	0.808	−0.084	0.885	0.841	−0.044	0.038	0.238	**+0.200**
NVILA-Lite-8B	0.911	0.846	−0.065	0.857	0.795	−0.062	0.019	0.095	+0.076
Qwen3-VL-30B	0.916	0.879	−0.037	0.910	0.927	+0.017	0.019	0.038	+0.019
Pixtral-12B	0.870	0.805	−0.065	0.894	0.783	−0.111	0.067	0.086	+0.019
Llama-3.2-Vision-11B	0.942	0.837	−0.105	0.841	0.802	−0.039	0.019	0.086	+0.067
Qwen3.6-27B	0.894	0.925	+0.031	0.825	0.854	+0.029	0.000	0.000	0.000
InternVL2.5-8B	0.860	0.823	−0.037	0.877	0.546	−0.331	0.095	0.067	−0.028

without structured prompting and retrieval context, the image modality yields substantially less verified information that is not already recoverable from the associated text.

NLI Novelty decreases for six of the seven models, corresponding to higher NLI Grounding against the retrieved document context. The average novelty reduction is 0.052. Thus, the structured configuration does not simply encourage models to generate more text unsupported by the retrieved context. Instead, it increases verified enrichment while generally producing captions more closely connected to document evidence.

VQA consistency decreases on average, from 0.870 to 0.793, with only two models improving. This trade-off is consistent with the changed caption task rather than necessarily indicating poorer visual behaviour. The no-prompt baseline typically produces short, generic descriptions containing few and relatively simple claims. In contrast, the structured two-phase prompt elicits additional contextual claims, making captions longer and more specific. This increases the number and complexity of claims exposed to atomic visual verification, which can lower the mean confirmation proportion even when captions contain more genuinely useful image-derived information.

`InternVL2.5-8B` is the principal exception to the overall pattern. It is the only model whose verified enrichment decreases under structured prompting, and it also shows the largest decline in VQA consistency. This result is most plausibly interpreted as sensitivity to the longer, context-heavy prompt configuration rather than a precision mismatch. Its lowest-ranking position in Ablation 2 is therefore consistent with the paired diagnostic result.

5.7.2.3 Per-PDF Breakdown

Table 5.8 isolates the granular per-document breakdown across the corpus for the top-three models selected via the multi-objective utility-scoring criteria.

Table 5.8: Ablation 2 per-document results for the top-3 selected models under the VidoRe retriever framework. NLI = NLI Grounding Score; VQA = mean of VQA score.

Model	Document	N	NLI	VQA	Enrichment
Mistral-Small-3.1-24B	PDF-1	25	0.274	0.717	0.400
	PDF-2	18	0.245	0.847	0.333
	PDF-3	19	0.219	0.914	0.368
	PDF-4	38	0.116	0.872	0.053
	PDF-5	5	0.077	0.930	0.000
NVILA-Lite-8B	PDF-1	25	0.241	0.752	0.200
	PDF-2	18	0.293	0.873	0.278
	PDF-3	19	0.158	0.772	0.000
	PDF-4	38	0.047	0.787	0.000
	PDF-5	5	0.026	0.880	0.000
Qwen3-VL-30B	PDF-1	25	0.145	0.941	0.000
	PDF-2	18	0.136	0.919	0.056
	PDF-3	19	0.196	0.905	0.158
	PDF-4	38	0.070	0.932	0.000
	PDF-5	5	0.041	0.920	0.000

5.7.2.4 Metric Analysis and Key Findings

The seven candidates show no single dominant configuration across quality, speed, and size:

- **Mistral-Small-3.1-24B** records the highest enrichment rate in the field (0.238) and strong NLI (0.808) and VQA (0.841) scores, but at the highest computational cost among the top performers (24.0B parameters, 61.2 s/image). Its enrichment is variable across documents, reaching 0.400 on PDF-1 and 0.368 on PDF-3, but falling to 0.000 on PDF-5.

- **NVILA-Lite-8B** records the fastest latency in the field (14.3 s/image) and the smallest parameter count jointly with InternVL2.5-8B (8.1B), but its enrichment rate (0.095) is substantially lower than Mistral's and is concentrated in PDF-2 (0.278) and PDF-1 (0.200), reaching 0.000 in the remaining three documents.

- **Qwen3-VL-30B** records the highest VQA score across all seven candidates (0.927, peaking at 0.941 on PDF-1) and the highest NLI score (0.879), but its enrichment rate is the lowest of the three leading candidates (0.038 overall), reaching 0.000 on PDF-4, PDF-1, and PDF-5. It is also the largest model evaluated (30.3B parameters).

- **The remaining candidates** each combine at least one substantial weakness in quality, latency, or enrichment relative to the three leading candidates, with Qwen3.6-27B recording zero verified enrichment despite the highest latency in the field (107.0 s/image), and InternVL2.5-8B recording the lowest VQA score overall (0.546).

Mistral-Small-3.1-24B, NVILA-Lite-8B, and Qwen3-VL-30B are carried forward as the three highest-ranked configurations for the downstream context-injection ablations. Their selection preserves distinct operational profiles: Mistral-Small-3.1-24B provides the strongest quality–enrichment profile; NVILA-Lite-8B provides the fastest and smallest configuration; and Qwen3-VL-30B provides the highest VQA score. The relative performance of these candidates under the unified quality–speed–size criterion is assessed later in the global cross-ablation comparison (Section 5.7.7).

The top-three reduction is a constrained multi-objective screening step rather than final model selection. Models were retained because they combined high composite performance with complementary efficiency, enrichment, and visual-grounding characteristics relevant to subsequent context-injection experiments. Models ranked below the retained set neither achieved a higher composite score nor provided an operational advantage.

5.7.3 Ablation 3: Parallel Context Injection

Ablation 3 evaluates the top-3 selected models under an summary-augmented context framework. Global report summaries (compiled sequentially via an isolated MapReduce architecture with a Qwen2-VL-7B backbone) and VidoRe-retrieved local text arrays are injected into the prompt as distinct, unblended structural boundaries. This configuration evaluates the model's capacity to cross-reference global thematic guidelines with localized physical evidence without preprocessing or fusing the two data streams. Together, Ablations 3 and 4 are designed to test whether varying how local (per-page retrieval) and global (whole-document summary) context are combined changes the top-3 ranking established in Ablation 2, or whether that ranking is robust across context-handling strategies.

5.7.3.1 Raw Evaluation Metrics

Table 5.9 details the performance metrics and processing runtimes for the top-three model configurations under the parallel injection framework.

Table 5.9: Ablation 3 evaluation metrics: Top-3 model configurations evaluated under unblended, parallel summary + context prompting structures.

Model	Published Params (B)	s/img	NLI Novelty	Comb VQA	Enrichment
Qwen3-VL-30B	30.3B	26.4	0.854	0.920	0.057
NVILA-Lite-8B	8.1B	17.5	0.826	0.832	0.086
Mistral-Small-3.1-24B	24.0B	58.0	0.840	0.822	0.114

5.7.3.2 Ablation 3 Findings and Analysis

The implementation of parallel abstractive context yields distinct model responses relative to Ablation 2:

- **Qwen3-VL-30B** records the highest raw quality signal in this round, with NLI Text-Non-Entailment of 0.854 and VQA of 0.920. Its verified enrichment rate increases relative to Ablation 2 (0.038 → 0.057). Its observed end-to-end latency was

26.4 s/image, compared with 51.8 s/image in Ablation 2; because these runs use different prompt and context configurations, this difference should be interpreted as configuration-level throughput rather than as an architecture-intrinsic speed change.

- **NVILA-Lite-8B** records the lowest raw quality of the three configurations in this round, and its enrichment rate decreases slightly relative to Ablation 2 (0.095 → 0.086). It records the lowest observed latency (17.5 s/image) and remains the smallest model in this three-model comparison.
- **Mistral-Small-3.1-24B** records the highest enrichment rate of the three configurations in this round (0.114). Its observed latency decreases slightly relative to Ablation 2 (61.2 → 58.0 s/image), but it remains the slowest configuration in this round and carries the largest parameter count among the top-3 candidates evaluated here (24.0B).

Having evaluated the fixed top-3 configuration under this unblended, parallel context format, the following ablation evaluates whether pre-fusing local and global context into a single blended passage changes model behaviour further. The relative standing of these three configurations under a unified quality–speed–size criterion is assessed later in the cross-ablation comparison (Section 5.7.4.3).

Having evaluated the fixed top-3 configuration under this unblended, parallel context format, the following ablation evaluates whether pre-fusing local and global context into a single blended passage changes this ranking further.

5.7.4 Ablation 4: Blended 70/30 Context Fusion

Ablation 4 evaluates a pre-fusion contextual strategy: VidoRe-retrieved localized data vectors (70%) and macro-level report summaries (30%) are combined into a single, unified string prior to injection. This experiment measures whether pre-fusing thematic guidance directly with local source blocks limits token noise, preventing the models from becoming disoriented by the conflicting prompt boundaries observed during parallel execution.

5.7.4.1 Raw Evaluation Metrics

Table 5.10 details the raw performance metrics and processing runtimes for the top-three model configurations under the 70/30 pre-fused context framework.

Table 5.10: Ablation 4 raw evaluation metrics: top-3 model configurations evaluated under unified, 70/30 pre-fused context and summary prompting sequences.

Model	Published Params (B)	s/img	NLI Novelty	Comb VQA	Enrichment
NVILA-Lite-8B	8.1B	16.7	0.783	0.840	0.181
Mistral-Small-3.1-24B	24.0B	67.7	0.796	0.882	0.124
Qwen3-VL-30B	30.3B	44.3	0.857	0.910	0.019

5.7.4.2 Ablation 4 Findings and Strategy Selection

The 70/30 blended context matrix alters the enrichment hierarchy seen in previous rounds:

- **NVILA-Lite-8B** reaches its highest verified enrichment rate across the context-injection ablations (0.181), while also recording the lowest observed latency in the round (16.7 s/image) and remaining the smallest of the three models.
- **Mistral-Small-3.1-24B** records the strongest VQA score in the round (0.882), but has the highest observed latency of the three models (67.7 s/image) and the largest parameter count among the top-3 (24.0B).
- **Qwen3-VL-30B** maintains a high VQA score (0.910), but its enrichment rate falls to 0.019, the lowest of the three configurations in this round. Within this three-model comparison, the pre-fused 70/30 context format is associated with lower verified enrichment for Qwen3-VL-30B than its parallel-context configuration in Ablation 3.

5.7.4.3 Global Context Strategy Cross-Comparison

Table 5.11 traces the development of the verified Enrichment Rate for the fixed top-3 models across Ablations 2, 3, and 4 to measure the exact impact of each context-handling method.

Table 5.11: Granular progression of verified Enrichment Rates across Ablations 2, 3, and 4 for the fixed top-3 models. $\Delta_{3\text{-}2}$ indicates the step change from the structured-prompt baseline to parallel summaries; $\Delta_{4\text{-}3}$ tracks the net change from parallel to blended context pre-fusion.

Model	Abl. 2 (Structured)	Abl. 3 (Unblended)	Abl. 4 (Blended)	$\Delta_{3\text{-}2}$	$\Delta_{4\text{-}3}$
Mistral-Small-3.1-24B	0.238	0.114	0.124	−0.124	+0.010
Qwen3-VL-30B	0.038	0.057	0.019	+0.019	−0.038
NVILA-Lite-8B	0.095	0.086	0.181	−0.009	**+0.095**

Across the full Ablation 2→4 progression, `NVILA-Lite-8B` is the only model of the three whose enrichment rate improves overall (net +0.086, from 0.095 to 0.181), with the sharpest single gain occurring at the parallel-to-blended transition (+0.095). Both `Mistral-Small-3.1-24B` (net −0.114) and `Qwen3-VL-30B` (net −0.019) end the progression lower than they started. `Mistral-Small-3.1-24B`'s processing speed fluctuates across the campaign (61.2 → 58.0 → 67.7 s/img), improving marginally under parallel injection before deteriorating again under blended fusion, though it remains among the slowest configurations at every stage regardless of context format.

5.7.4.4 Model Selection Weighted Multi-Criteria Ranking (Ablations 2–4)

To formally select the model and context strategy carried forward into the prompt-engineering (Ablation 5) and decoding-level (Ablation 6) refinement tracks, a weighted

multi-criteria ranking was conducted across all nine configurations executed in Ablations 2 through 4: the three candidate models (Mistral-Small-3.1-24B, NVILA-Lite-8B, Qwen3-VL-30B) under each of the three context strategies (structured prompt, parallel injection, blended 70/30 fusion). This comparison uses the weighting scheme described in Section 5.4.5.

Table 5.12 reports the resulting ranking. Because the raw metrics underlying each entry were already presented in Sections 5.7.2 through 5.7.4, only the normalized composite scores are shown here.

Table 5.12: Model selection weighted multi-criteria ranking across all nine model–context configurations executed in Ablations 2 through 4. This ranking determines which single configuration is carried forward into Ablations 5 and 6.

Rank	Model	Ablation	Q_raw	Q_norm	S_norm	Z_norm	Composite
1	**NVILA-Lite-8B**	**Abl. 4**	0.601	0.638	0.955	1.000	**0.7718**
2	Mistral-Small-3.1-24B	Abl. 2	0.625	1.000	0.120	0.284	0.6727
3	Qwen3-VL-30B	Abl. 3	0.608	0.753	0.773	0.000	0.6448
4	Qwen3-VL-30B	Abl. 2	0.609	0.757	0.298	0.000	0.5287
5	NVILA-Lite-8B	Abl. 3	0.570	0.179	0.940	1.000	0.4924
6	Mistral-Small-3.1-24B	Abl. 4	0.604	0.683	0.000	0.284	0.4523
7	Qwen3-VL-30B	Abl. 4	0.590	0.485	0.439	0.000	0.4004
8	NVILA-Lite-8B	Abl. 2	0.558	0.000	1.000	1.000	0.4000
9	Mistral-Small-3.1-24B	Abl. 3	0.577	0.290	0.181	0.284	0.2617

The ranking shows that the preferred configuration is determined by the quality–efficiency trade-off rather than by raw quality alone. Although Mistral-Small-3.1-24B under Ablation 2 achieves the highest raw quality score, its substantially slower inference and larger parameter count reduce its overall deployment suitability. Conversely, the Ablation 4 configuration of NVILA-Lite-8B combines competitive quality with consistently strong speed and size efficiency, producing the highest composite score across the nine evaluated model–context configurations.

This result also indicates that the blended 70/30 context-fusion strategy is more suitable for NVILA-Lite-8B than its structured-prompt and parallel-context alternatives within the evaluated configuration set. Qwen3-VL-30B remains competitive on quality and throughput under parallel context injection, but its larger parameter footprint prevents it from matching the overall quality–speed–size balance of NVILA-Lite-8B.

Accordingly, **NVILA-Lite-8B with the Ablation 4 blended** 70/30 **context-fusion strategy** is selected as the single base configuration carried forward into the prompt-engineering (Ablation 5) and decoding-level (Ablation 6) refinement tracks.

5.7.5 Ablation 5: Prompt Engineering and Hallucination Control

Building upon the weighted multi-criteria selected baseline configuration from Ablation 4, Ablation 5 evaluates five advanced prompt-engineering methodologies, all applied to `NVILA-Lite-8B` with the 70/30 blended context format. The six prompt configurations are defined in Section 4.4.6.

5.7.5.1 Raw Evaluation Metrics

Table 5.13 details the raw performance metrics and processing runtimes for the baseline and five prompt-engineering configurations.

Table 5.13: Ablation 5 raw evaluation metrics: comparison of a baseline against advanced prompt engineering and visual guardrail strategies applied to the NVILA-Lite-8B architecture.

Config	Prompt Strategy	s/img	NLI Novelty	Comb VQA	Enrich
A5-P4	Internal Self-Correction	12.7	0.805	0.861	0.114
A5-P3	Counterfactual Contrast	13.8	0.771	0.828	**0.219**
A5-P0	Baseline (no prompt engineering)	16.7	0.783	0.840	0.181
A5-P5	Domain Expert Persona	16.0	0.803	0.837	0.143
A5-P2	Verification & Refusal Gate	16.1	0.777	0.815	0.152
A5-P1	Visual Primitive Guardrail	17.4	0.790	0.820	0.086

5.7.5.2 Prompt Strategic Analysis and Findings

Including the unmodified baseline (A5-P0) in this round shows that the prompt strategies produce different quality–throughput trade-offs, all under the same NVILA-Lite-8B architecture:

- **Counterfactual Contrast (A5-P3)** records the highest verified enrichment rate of the six configurations (0.219) and a strong combined VQA score (0.828), at a latency of 13.8 s/image. Explicitly logging visually absent items relative to context (*Bilan d'Absence*) is associated with higher enrichment in this ablation, although the design does not isolate which prompt element produces this effect.
- **Internal Self-Correction (A5-P4)** records the lowest observed latency in the round (12.7 s/image) and the highest combined VQA score (0.861), with an enrichment rate of 0.114, lower than A5-P3.
- The remaining prompt strategies (Baseline A5-P0, Domain Expert Persona A5-P5, Verification & Refusal Gate A5-P2, and Visual Primitive Guardrail A5-P1) did not exceed A5-P3 or A5-P4 on either enrichment or latency, with Visual Primitive Guardrail recording the lowest enrichment rate of the six configurations (0.086) and the highest observed latency in the round (17.4 s/image).

The six configurations expose a trade-off between verified enrichment, visual consistency, and observed throughput. Within the six Ablation 5 configurations, Within the six Ablation 5 configurations, Counterfactual Contrast (A5-P3) achieves the strongest raw quality and verified enrichment, whereas Internal Self-Correction (A5-P4) provides the lowest observed latency and the strongest combined VQA score. All six configurations are retained for the global cross-ablation weighted multi-criteria comparison in Section 5.7.7.

5.7.6 Ablation 6: Contrastive Decoding

To investigate the theoretical limits of prompt-based context engineering, Ablation 6 pivots away from textual prompting strategies to evaluate token-generation level logit filtering, applied to the selected `NVILA-Lite-8B` configuration. This track evaluates Instruction Contrastive Decoding (ICD) and Visual Contrastive Decoding (VCD), each at a mild and a strong contrastive intensity setting, against the 70/30 blended context baseline established in Ablation 4. By modifying the decoder's soft mathematical properties during autoregressive token selection, these frameworks seek to mathematically suppress language-prior-driven text expectations or under-grounded visual guessing by subtracting perturbed logit distributions from the baseline distribution. The ICD and VCD procedures are defined in Section 4.4.7.

5.7.6.1 Raw Evaluation Metrics

Table 5.14 presents the raw performance metrics and processing latencies for the four contrastive decoding configurations.

Table 5.14: Ablation 6 raw evaluation metrics: performance metrics for logit-level Instruction Contrastive Decoding (ICD) and Visual Contrastive Decoding (VCD) applied to NVILA-Lite-8B, at mild ($\alpha = 0.5$) and strong ($\alpha = 1.0$) settings.

Configuration	s/img	NLI Novelty	Comb VQA	Enrichment
VCD mild ($\alpha = 0.5$, $t = 500$)	29.3	0.805	0.835	0.171
ICD mild ($\alpha = 0.5$)	20.6	0.818	0.815	0.124
VCD strong ($\alpha = 1.0$, $t = 500$)	41.6	0.826	0.817	0.105
ICD strong ($\alpha = 1.0$)	19.6	0.818	0.779	0.086

5.7.6.2 Decoder Modification Findings and Analysis

With only `NVILA-Lite-8B` configurations under evaluation, the results isolate the effect of contrastive decoding mechanism and intensity independent of model architecture:

- **VCD mild** records the highest raw quality of the four configurations, with a combined VQA score of 0.835 and an enrichment rate of 0.171. This enrichment figure sits close to the best prompt-engineering result from Ablation 5 (A5-P3, 0.219), indicating that a gently-tuned visual contrastive decoder can approach, though not exceed, prompt-engineering-driven enrichment gains, without requiring any prompt redesign.
- **ICD is consistently faster, but lower quality, than VCD at matching intensity:** ICD mild (20.6 s/img) and ICD strong (19.6 s/img) are both faster than their VCD counterparts (29.3 s/img and 41.6 s/img respectively), since VCD's dual-image contrastive pass carries a larger compute overhead than ICD's dual-prompt approach. However, at both intensity levels, ICD trails VCD on combined VQA (0.815 vs. 0.835 at mild; 0.779 vs. 0.817 at strong) and on enrichment (0.124 vs.

0.171 at mild; 0.086 vs. 0.105 at strong), so ICD's speed advantage comes at a consistent quality cost.

- **The intensity parameter (α) penalty:** Increasing the contrastive intensity from mild ($\alpha = 0.5$) to strong ($\alpha = 1.0$) worsens enrichment for both mechanisms: VCD falls from 0.171 to 0.105, and ICD falls from 0.124 to 0.086, indicating that aggressive logit-level filtering increasingly restricts the vocabulary space available for descriptive, relational content regardless of which contrastive mechanism is used.
- **Processing cost:** All four configurations record higher observed end-to-end latency than the standard single-pass NVILA-Lite-8B configuration in Ablation 4 (16.7 s/image), since both ICD and VCD require dual-prompt or dual-image forward passes to compute contrastive logit adjustments at each generation step.

5.7.7 Cross-Ablation Synthesis and Model Selection

To support final deployment selection, a single global weighted multi-criteria comparison was conducted across every configuration executed in Ablations 2 through 6, using the scoring methodology and weighting scheme described in Section 5.4.5. Ablation 1 (Section 5.7.1) emphasized the use of structured prompting and retrieval context and is not included in this comparison.

Model selection proceeded in two stages. First, a weighted multi-criteria comparison was conducted across the nine model–context configurations executed in Ablations 2 through 4 (Section 5.7.4.4), from which NVILA-Lite-8B under the blended 70/30 context-fusion strategy emerged as the leading configuration. This configuration was then carried forward as the sole base model for the prompt-engineering (Ablation 5) and decoding-level (Ablation 6). The global comparison presented here pools the outcomes of both stages: the nine configurations from Ablations 2 through 4, the five prompt-engineering variants applied to NVILA-Lite-8B in Ablation 5, and the four contrastive-decoding configurations applied in Ablation 6, for a combined pool of 18 configurations.

This global comparison differs from the two earlier stages in its normalization scope. In each earlier stage, quality, throughput, and published parameter count were normalized using the minimum and maximum values within that stage's own candidate set, so those scores are not comparable across stages. Here, the same three metrics are normalized once across all 18 configurations, so every executed configuration is measured on a common scale and can be compared directly against every other.

The global candidate pool was evaluated using the same weighting scheme applied throughout the preceding ablations:

$$\text{Quality} = 0.60, \qquad \text{Speed} = 0.25, \qquad \text{Size} = 0.15,$$

where the quality component assigns weights of VQA = 0.50, Enrichment = 0.35, and NLI Novelty = 0.15. The size component uses published total architecture parameter counts, inverted during min–max normalization so that smaller models receive higher size scores; the speed component uses observed end-to-end latency in seconds per image, inverted so that faster configurations receive higher scores.

5.7.7.1 Global Multi-Criteria Results

Table 5.15 reports the raw quality, throughput, and published-parameter values for all 18 configurations. Table 5.16 reports the final composite ranking.

Table 5.15: Global cross-ablation comparison : all 18 configurations executed across Ablations 2 through 6.

Rank	Model	Ablation	Params (B)	s/img	NLI Novelty	Comb VQA	Enrich-ment
1	**NVILA-Lite-8B**	**Abl. 5(P3)**	8.1	13.8	0.771	0.828	0.219
2	NVILA-Lite-8B	Abl. 4	8.1	16.7	0.783	0.840	0.181
3	NVILA-Lite-8B	Abl. 5 (P4)	8.1	12.7	0.805	0.861	0.114
4	NVILA-Lite-8B	Abl. 6 (VCD 0.5)	8.1	29.3	0.805	0.835	0.171
5	NVILA-Lite-8B	Abl. 5 (P5)	8.1	16.0	0.803	0.837	0.143
6	Mistral-Small-3.1-24B	Abl. 2	24.0	61.2	0.808	0.841	0.238
7	Qwen3-VL-30B	Abl. 3	30.3	26.4	0.854	0.920	0.057
8	NVILA-Lite-8B	Abl. 5 (P2)	8.1	16.1	0.777	0.815	0.152
9	NVILA-Lite-8B	Abl. 6 (ICD 0.5)	8.1	20.6	0.818	0.815	0.124
10	NVILA-Lite-8B	Abl. 3	8.1	17.5	0.826	0.832	0.086
11	Qwen3-VL-30B	Abl. 2	30.3	51.8	0.879	0.927	0.038
12	NVILA-Lite-8B	Abl. 2	8.1	14.3	0.846	0.795	0.095
13	NVILA-Lite-8B	Abl. 5 (P1)	8.1	17.4	0.790	0.820	0.086
14	Mistral-Small-3.1-24B	Abl. 4	24.0	67.7	0.796	0.882	0.124
15	NVILA-Lite-8B	Abl. 6 (VCD 1.0)	8.1	41.6	0.826	0.817	0.105
16	Qwen3-VL-30B	Abl. 4	30.3	44.3	0.857	0.910	0.019
17	NVILA-Lite-8B	Abl. 6 (ICD 1.0)	8.1	19.6	0.818	0.779	0.086
18	Mistral-Small-3.1-24B	Abl. 3	24.0	58.0	0.840	0.822	0.114

Table 5.16: Global cross-ablation comparison,weighted scoring computed over the full 18-configuration candidate pool.

Rank	Model	Ablation	Q raw	Q norm	S norm	Z norm	Composite score
1	**NVILA-Lite-8B**	**Abl. 5(P3)**	0.606	0.776	0.906	1.000	**0.8423**
2	NVILA-Lite-8B	Abl. 4	0.601	0.705	0.928	1.000	**0.8051**
3	NVILA-Lite-8B	Abl. 5 (P4)	0.591	0.590	1.000	1.000	**0.7537**
4	NVILA-Lite-8B	Abl. 6 (VCD 0.5)	0.598	0.676	0.698	1.000	**0.7299**
5	NVILA-Lite-8B	Abl. 5 (P5)	0.589	0.564	0.940	1.000	**0.7230**
6	Mistral-Small-3.1-24B	Abl. 2	0.625	1.000	0.117	0.284	**0.6718**
7	Qwen3-VL-30B	Abl. 3	0.608	0.799	0.750	0.000	**0.6667**
8	NVILA-Lite-8B	Abl. 5 (P2)	0.577	0.424	0.939	1.000	**0.6394**
9	NVILA-Lite-8B	Abl. 6 (ICD 0.5)	0.573	0.375	0.856	1.000	**0.5889**
10	NVILA-Lite-8B	Abl. 3	0.570	0.331	0.913	1.000	**0.5769**
11	Qwen3-VL-30B	Abl. 2	0.609	0.802	0.290	0.000	**0.5535**
12	NVILA-Lite-8B	Abl. 2	0.558	0.185	0.971	1.000	**0.5039**
13	NVILA-Lite-8B	Abl. 5 (P1)	0.558	0.193	0.915	1.000	**0.4944**
14	Mistral-Small-3.1-24B	Abl. 4	0.604	0.742	0.000	0.284	**0.4876**
15	NVILA-Lite-8B	Abl. 6 (VCD 1.0)	0.569	0.322	0.475	1.000	**0.4619**
16	Qwen3-VL-30B	Abl. 4	0.590	0.580	0.426	0.000	**0.4545**
17	NVILA-Lite-8B	Abl. 6 (ICD 1.0)	0.542	0.000	0.874	1.000	**0.3685**
18	Mistral-Small-3.1-24B	Abl. 3	0.577	0.421	0.176	0.284	**0.3393**

5.7.7.2 Global Winner and Strategic Conclusion

Within the full 18-configuration global comparison and the specified quality–throughput–size weighting scheme, **NVILA-Lite-8B with Counterfactual Contrast prompt engineering (Ablation 5, P3)** obtains the highest final composite score (0.8423). It combines the second-highest raw quality in the pool ($Q_{\text{raw}} = 0.606$), strong observed throughput (13.8 s/image; $S_{\text{norm}} = 0.906$), and the smallest published parameter count (8.1B; $Z_{\text{norm}} = 1.000$).

The Ablation 4 blended 70/30 context-fusion baseline ranks second ($Q_{\text{raw}} = 0.601$, composite score 0.8051). Counterfactual Contrast improves on this baseline primarily through higher verified enrichment (0.219 versus 0.181) and higher NLI Novelty, at the cost of a marginal latency increase. Internal Self-Correction (Ablation 5, P4) is the fastest configuration in the pool (12.7 s/image), but its lower raw quality and enrichment place it third under the global weighting scheme.

Mistral-Small-3.1-24B under Ablation 2 achieves the highest raw quality ($Q_{\text{raw}} = 0.625$) and enrichment rate (0.238), but its slower inference and larger parameter count place it sixth overall. The Qwen3-VL-30B configurations retain strong VQA scores but receive no size advantage in the global comparison because they have the largest published parameter count in the pool.

The final selection follows the two-stage experimental design: Ablations 2 through 4 first selected the `NVILA-Lite-8B` blended-context configuration as the base system, after which Ablations 5 and 6 evaluated prompt-engineering and decoding refinements of that system. Under the global comparison across all 18 executed configurations, **NVILA-Lite-8B with Counterfactual Contrast prompt engineering** ranks first under the specified weighted quality–throughput–size composite score and is therefore carried forward into Pipeline 4.

5.8 Pipeline 4: Baseline vs. Fine-Tuned Enrichment

Pipeline 4 was evaluated on a corpus of five heterogeneous river-restoration documents, spanning technical fiches, internship reports, species-monitoring dossiers, and historical cartographic studies, under the controlled ablation protocol described in Section 4.5.8: an identical extraction, serialization, and entity-resolution pipeline executed once with the base Qwen2.5-14B-Instruct model[56] (baseline) and once with the LoRA fine-tuned checkpoint (fine-tuned), differing only in the model weights. Three quantitative metrics were computed for each condition on every document: Schema Conformance Rate (SCR), Average Node Degree Shift ($\Delta\bar{d}$), and Cross-Modal Link Density (CMLD). In addition, an Intrinsic Triple Precision (P_T) audit protocol was prepared but not executed within the scope of this thesis, as defined in Section 4.5.9.

5.8.1 Triple and Entity-Level Enrichment

Table 5.17 and Table 5.18 report RDF triple and entity counts under each condition. The text and image columns refer to independently generated source graphs, whereas the merged column refers to the final RDF graph after cross-modal entity resolution. During merging, duplicate and alias entity nodes are folded into canonical URIs; consequently,

triples that become identical after this resolution are retained only once in the final graph. Merged-triple counts are therefore not expected to equal the arithmetic sum of the text- and image-derived source-graph counts.

Despite this expected cross-modal overlap, image-derived processing provides net structural enrichment for every document: each merged graph contains more RDF triples than its corresponding text-only graph, and the image pass introduces additional entities beyond those extracted from text. The results therefore indicate both complementary image-derived information and agreement between modalities where independently extracted facts resolve to the same canonical graph representation.

Table 5.17: Baseline pipeline: source-graph and final merged-graph triple counts, together with entity counts, per document. Merged triples are deduplicated after cross-modal entity resolution and therefore are not expected to equal the sum of text and image triples. Document abbreviations (PDF-1–PDF-5) are defined in Section 5.2.

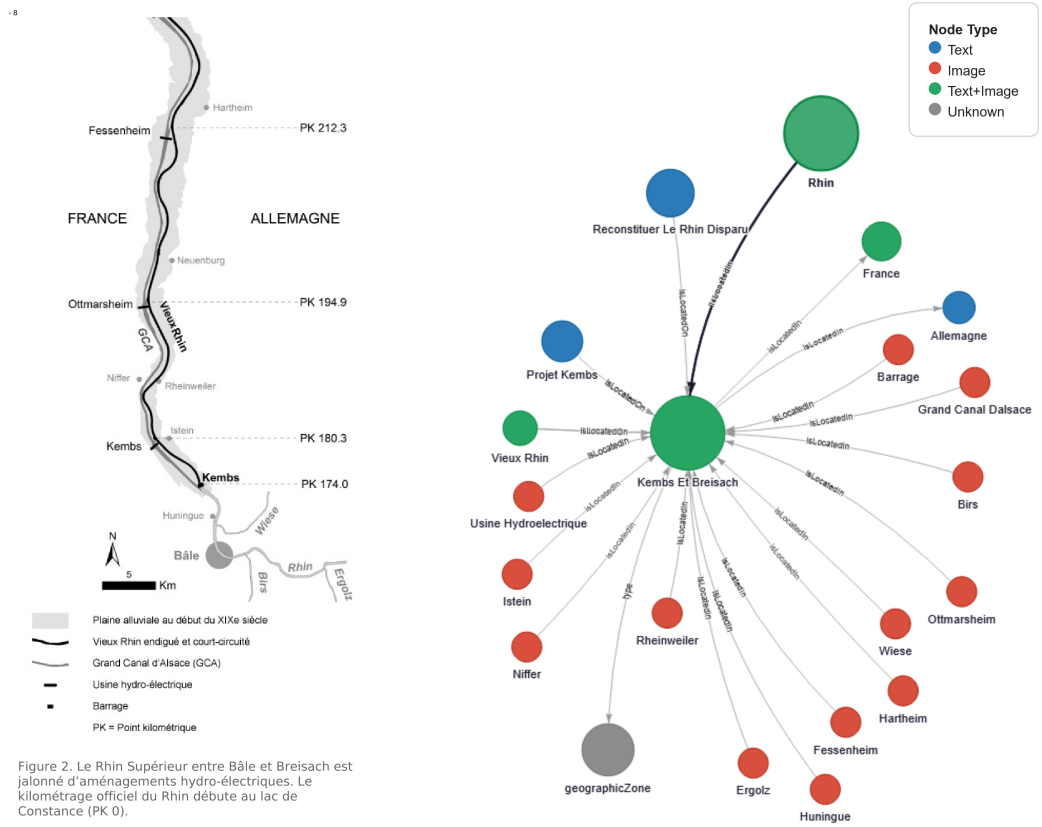
Document	Text Triples	Image Triples	Merged Triples	New Entities (Images)	Cross-Modal Entities
PDF-1	946	694	1,481	26	30
PDF-2	3,248	795	3,798	58	31
PDF-3	1,441	632	1,892	32	36
PDF-4	4,014	1,550	5,143	90	68
PDF-5	507	334	744	21	22
Corpus total	**10,156**	**4,005**	**13,058**	**227**	**187**

Table 5.18: Fine-tuned pipeline: source-graph and final merged-graph triple counts, together with entity counts, per document. Merged triples are deduplicated after cross-modal entity resolution and therefore are not expected to equal the sum of text and image triples. Document abbreviations (PDF-1–PDF-5) are defined in Section 5.6.1.

Document	Text Triples	Image Triples	Merged Triples	New Entities (Images)	Cross-Modal Entities
PDF-1	643	463	973	11	23
PDF-2	1,760	187	1,861	17	7
PDF-3	803	361	1,030	15	21
PDF-4	2,366	833	3,009	67	24
PDF-5	424	182	549	14	9
Corpus total	**5,996**	**2,026**	**7,422**	**124**	**84**

A notable observation is that the fine-tuned pipeline produces substantially fewer total triples than the baseline across every document (7,422 vs. 13,058 corpus-wide, a 43.2% reduction). This reduction is accompanied by a proportionally similar drop in new image-derived entities (124 vs. 227, −45.4%) and cross-modal linked entities (84 vs. 187, −55.1%), indicating that the volume decrease is broadly uniform across triple types rather than con-

centrated in any single category. Section 5.8.2 interprets this reduction jointly with the schema-conformance results below: the smaller output is compatible with reduced speculative or off-ontology generation, but may also reflect increased conservatism or omission of valid facts. The available structural metrics cannot distinguish these explanations because triple-level factual precision was not measured (Section 5.8.2.2). Figure 5.1 provides a concrete PDF-5 example of image-derived entities incorporated into the merged graph.



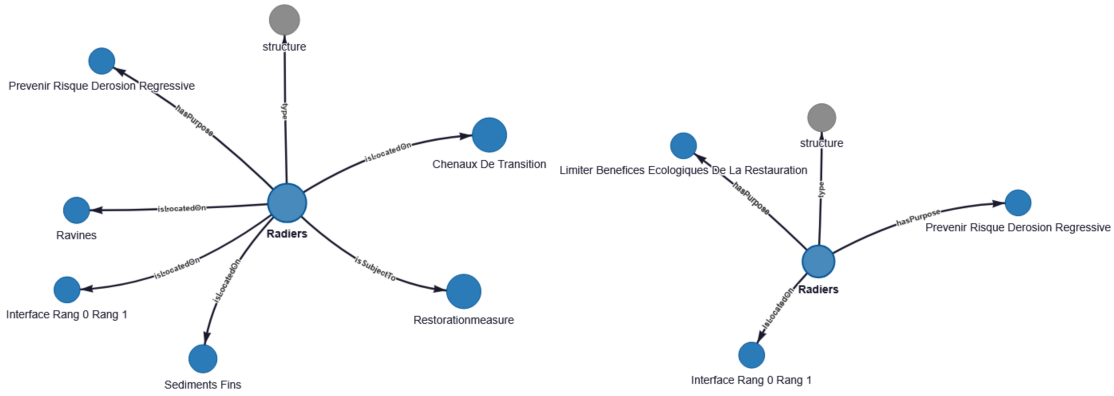
(a) Source figure (PDF-5): the Rhine between Basel and Breisach.

(b) Corresponding knowledge-graph subgraph, centered on the "Kembs et Breisach" entity.

Figure 5.1: Example of cross-modal enrichment on a single source document (PDF-5). In this subgraph, the image-caption extraction introduces entities such as surrounding towns, tributaries, and hydraulic infrastructure that do not appear in the retrieved text, illustrating in concrete terms the new-entity and cross-modal-entity counts reported for PDF-5 in Table 5.17.

5.8.2 Schema Conformance and Cross-Modal Enrichment

Figure 5.2 provides a source-verified local comparison of graph construction under the baseline and fine-tuned configurations. It is intended as an illustrative structural example rather than a corpus-level measure of factual accuracy. The two panels are centred on the same radiers entity and show how fine-tuning changes the retained local neighborhood.



(a) Baseline local subgraph for the radiers entity.

(b) Fine-tuned local subgraph for the same entity.

Figure 5.2: Illustrative baseline-versus-fine-tuned local graph comparison for the shared radiers entity. The source passage states that radiers are placed at the rang 0/rang 1 interface to prevent regressive erosion. The fine-tuned graph retains these two source-supported relations while producing a smaller local neighborhood. The baseline graph additionally links the entity to downstream erosion consequences, including ravines and fine-sediment loss, through `isLocatedOn`. This qualitative example illustrates improved local structural selectivity; it does not ensure factual accuracy.

Table 5.19 compares the three graph-quality metrics between the baseline and fine-tuned conditions for each document, and Table 5.20 summarizes the corpus-level aggregates.

Table 5.19: Per-document comparison of SCR(5.4.6.1), Average Node Degree Shift (Section 5.4.6.2), and CMLD(5.4.6.3 between baseline and fine-tuned pipelines. Document identifiers are defined in Section 5.2. BL = Baseline; FN = Fine-tuned.

Document	**SCR**		$\Delta\bar{d}$		**CMLD**	
	BL	**FN**	**BL**	**FN**	**BL**	**FN**
PDF-1	56.65%	**90.42%**	+1.6995	**+2.6862**	52.49%	**78.09%**
PDF-2	79.37%	**84.35%**	+0.4421	+0.0602	24.67%	**39.62%**
PDF-3	72.04%	**90.37%**	+0.6178	+0.2885	56.51%	**73.36%**
PDF-4	72.56%	**77.78%**	+0.6246	+0.3897	52.49%	**55.24%**
PDF-5	94.41%	**90.91%**[2]	+0.5860	+0.1140	68.42%	**66.00%**[3]

Fine-tuning improves Schema Conformance Rate (SCR) on four of five documents and at corpus level. This is the primary quantitative evidence that LoRA fine-tuning reduces the rate at which the model emits predicates outside the authorized TetraOnto vocabulary. The sole exception is PDF-5, where the baseline slightly exceeds the fine-tuned model. As the smallest document by triple count, its SCR is more sensitive to a

[3]On this document, the baseline slightly exceeds the fine-tuned model on SCR and CMLD; see discussion below.

Table 5.20: Corpus-level aggregate comparison, baseline vs. fine-tuned pipeline.

Metric	Baseline	Fine-tuned	Change
Total merged triples	13,058	7,422	−43.2%
Total new entities (images)	227	124	−45.4%
Total cross-modal entities	187	84	−55.1%
Average SCR	75.01%	**86.77%**	**+11.76 pp**
Average $\Delta\bar{d}$	+0.7940	+0.7077	−0.0863
Average CMLD	50.92%	**62.46%**	**+11.54 pp**

small number of predicate classifications and should be interpreted alongside the corpus-level trend. Residual non-conformance may also reflect gaps between ontology vocabulary and evaluation-corpus visual content, rather than model behaviour alone (Section 4.5).

Cross-Modal Link Density (CMLD) follows the same overall pattern, increasing for the fine-tuned condition on four documents and at corpus level. This indicates that a larger proportion of retained domain-relation output is associated with entities shared across text and image modalities. However, CMLD is computed over the smaller fine-tuned graph; part of the increase may therefore result from the reduced denominator rather than a higher absolute number of cross-modal links. It should consequently be interpreted as a relative cross-modal integration measure, not evidence that fine-tuning increases the total number of extracted cross-modal facts.

Average Node Degree Shift remains positive for every document and under both conditions, indicating increased URI-level graph density after image-derived content is merged with the text-based graph. The corpus-level degree shifts remain comparable despite the substantially smaller fine-tuned graph. The fine-tuned configuration produces a larger shift for PDF-1, whereas the baseline produces larger shifts for the remaining documents. Thus, the structural effect of image integration persists under fine-tuning but is not uniform across document types.

The fine-tuned checkpoint was trained on a synthetically generated, schema-filtered instruction corpus (Section 4.5.7). The filtering enforced valid JSON structure, non-empty triple fields, and membership of retained predicates in the TetraOnto vocabulary; it did not factually validate extracted triples. The following probe therefore evaluates whether the intended predicate-compliance behaviour transfers to previously unseen real image captions.

5.8.2.1 Preliminary Predicate-Compliance Probe

Before the full five-document extraction, a controlled probe compared baseline and fine-tuned model outputs on three previously unseen image captions sampled from each document (15 captions total). Generated predicates were automatically checked against the ontology whitelist. Table 5.21 reports the results.

This small-scale probe more than doubled the on-ontology predicate rate under fine-tuning (17.9% to 43.5%) and is directionally consistent with the SCR results in Table 5.20, although absolute rates differ between experiments. The probe measures raw model out-

Table 5.21: Preliminary predicate-compliance probe on 15 unseen image captions (3 per document).

Condition	Triples Generated	Out-of-Ontology	On-Ontology Rate
Baseline	56	46	17.9%
Fine-tuned	62	35	43.5%

put before downstream whitelist-and-demotion filtering, whereas SCR is computed after the deterministic serialization pipeline processes generated predicates (Section 4.5.5). It therefore provides an auxiliary measure of predicate-vocabulary compliance at generation time, while corpus-level SCR characterizes predicate conformance in the resulting graph after pipeline safeguards. The probe does not assess whether individual triples are factually supported by their source captions or passages and should not be interpreted as a measure of hallucination rate or intrinsic triple precision.

5.8.2.2 Proposed Intrinsic Triple Precision Audit and Remaining Limitations

For a future source-grounded factual-precision audit, stratified samples of up to 50 ontology-conformant triples per document and condition were extracted into `precision_sample_baseline.txt` and `precision_sample_finetuned.txt`. These samples were prepared for manual review against the originating source passage or verified image caption; however, no completed annotation set was produced within this report.

The structural metrics reported in this section establish improved predicate-vocabulary conformance under fine-tuning, with changes in graph connectivity and cross-modal link composition. They do not establish whether the fine-tuned model produces a higher proportion of factually supported triples. The observed reduction in raw triple volume may reflect reduced speculative extraction, increased conservatism, omission of valid facts, or a combination of these effects. Distinguishing these explanations requires the planned source-grounded manual annotation audit. The preliminary predicate-compliance probe in Section 5.8.2.1 should therefore be interpreted only as evidence of ontology-vocabulary compliance in raw model output, not factual correctness or reduced hallucination.

5.8.3 Empirical Synthesis and Deployment Implications

Evaluation across all five document pipelines provides an empirical roadmap for multimodal system optimization. The findings show that logit-level contrastive decoding introduced higher end-to-end latency than the strongest prompt-engineering configuration and could restrict the diversity of generated technical vocabulary. Across evaluated configurations, the results make trade-offs among output structure, retrieval and captioning behaviour, and graph-level ontology conformance explicit. For Pipeline 4, LoRA fine-tuning improved measured predicate-vocabulary conformance and altered graph connectivity and cross-modal link composition, while triple-level factual precision remains unmeasured. This quantitative framework completes the experimental analysis. Chapter 6 synthesizes the findings and limitations of the evaluated engineering configurations and assesses remaining systemic limitations.

Chapter 6

Conclusion and Perspectives

This thesis has detailed the architectural design, algorithmic implementation, and multi-criteria evaluation of a layout-aware multi-modal document intelligence framework. Developed as a core module for the **TETRA project** digital toolbox, the system integrates textual records and embedded visual assets into a common extraction and knowledge-graph workflow within French hydro-environmental engineering reports along the Rhine catchment. The resulting workflow converts non-linear PDF content into a machine-interpretable representation for information discovery in the water-resource management domain.

Synthesis of Experimental Findings and Results

This section answers the research questions introduced in Section 1 using the evaluation results reported in Chapter 5.

- **RQ1: Layout-aware extraction.**

  The layout-aware rasterization and `DocLayout-YOLOv10` procedure produced coherent visual crops in the qualitative and quantitative Pipeline 1 evaluations. The structural filtering results show that a substantial proportion of raw document segments consisted of repeated boilerplate, captions, or non-informative graphic regions, supporting the use of layout-aware extraction and multi-stage filtering before retrieval and captioning. Within the evaluated corpus, RQ1 is therefore answered positively: the procedure produced figure-level assets suitable for downstream processing while reducing structural document noise.

- **RQ2: Cross-modal retrieval.**

  The retrieval benchmark compared six candidate configurations on the manually annotated image–text associations. The document-aware single-vector `vdr-2b-multi-v1` configuration achieved the highest macro-nDCG@5 (0.5858) and led on three of the four reported ranking metrics. Within the internally annotated evaluation corpus, RQ2 is therefore answered by selecting VDR as the retrieval configuration used by Pipeline 3. This result remains bounded by the internally constructed ground truth and the configuration differences documented for the CLIP baseline.

- **RQ3: Grounded image captioning.**

  The Pipeline 3 ablations showed that the tested two-phase prompting configuration was associated with higher verified enrichment than the no-prompt baseline, while VQA-based visual consistency remained sensitive to caption complexity and atomic-claim count. The global comparison across all 18 executed configurations

selected `nvila-lite-8b` with the Counterfactual Contrast prompt strategy from Ablation 5. This configuration obtained the highest composite score (0.8423), combining $Q_{\text{raw}} = 0.606$, an observed end-to-end latency of 13.8 s/image, and an 8.1B published parameter count. RQ3 is therefore answered within the tested configurations: retrieval-grounded, pixel-first prompting combined with Counterfactual Contrast produced the strongest measured quality–throughput–size trade-off, while further factual-grounding evaluation remains necessary.

- **RQ4: Multi-modal knowledge-graph enrichment.**

  The controlled Pipeline 4 comparison showed that image-derived captions added entities and triples beyond the text-only graph, and that both baseline and fine-tuned conditions produced positive Average Node Degree Shifts. Predicate-focused LoRA fine-tuning increased the corpus-level Schema Conformance Rate from 75.01% to 86.77% (+11.76 percentage points) and Cross-Modal Link Density from 50.92% to 62.46% (+11.54 percentage points). The fine-tuned pipeline generated fewer total triples (7,422 versus 13,058 for the baseline), a reduction consistent with, but not sufficient to prove, lower speculative extraction. RQ4 is therefore answered at the structural and schema-conformance level; factual triple precision remains pending the manual review protocol.

Limitations and Systemic Perspectives

While the experimental study evaluates the principal extraction, retrieval, captioning, and graph-construction layers, several limitations remain open for continued research and refinement:

1. **Baseline Control Interpretation:** Ablation 1 was used as a diagnostic no-prompt control rather than as part of the deployment-selection process. Its captions are generally shorter and contain fewer claims than the structured outputs, which can inflate per-claim visual verification scores. The trade-off between enrichment and VQA consistency should therefore be interpreted in the context of differing caption complexity rather than as an absolute measure of model reliability. This caption-length confound remains relevant when comparing grounding and enrichment across configurations.

2. **Retrieval Ground-Truth Verification Bounds:** The 105 ground-truth image–text associations used for the Pipeline 2 evaluation were annotated internally by the project team rather than by an independent domain expert. The reported Recall@5, Precision@1, MRR, and nDCG@5 results should therefore be interpreted against this internally annotated ground truth. Independent validation remains necessary before the retrieval benchmark can be considered final.

3. **Factual Integrity and Triple Auditing:** Pipeline 4 applies deterministic predicate and class whitelisting during serialization, and LoRA fine-tuning increases Schema Conformance Rate from 75.01% to 86.77%. However, vocabulary-level conformance does not establish factual grounding, since a triple can use a valid predicate while asserting unsupported content. A stratified Intrinsic Triple Precision

audit protocol has been prepared, with samples of up to 50 triples per document and condition, but the manual annotation pass remains incomplete. Consequently, the reported Pipeline 4 results establish improved schema conformance and changes in graph structure, not improved factual triple precision. In addition, the fine-tuning data supervise predicate conformance but not `subject_type`; class-resolution accuracy at inference therefore remains dependent on the base model's zero-shot capability.

Future work involves prioritizing independent validation of the retrieval ground truth, completion of the manual precision audit across the full ten-file sample set (five reports across two pipeline conditions), and extension of the fine-tuning dataset to include class-typing supervision. Additional work may also investigate whether structured graph constraints can inform subsequent visual extraction or verification steps.

Teamwork and Institutional Collaboration

This research was conducted within the TETRA project, involving Fraunhofer IOSB, the ICube laboratory, SEBA, and Thales. The work benefited from discussions with team members working on database engineering, computer vision, and knowledge representation. In particular, collaboration supported the preparation of the TetraOnto schema, the alignment of layout coordinates with downstream processing, and the development of the four-stage architecture. The resulting codebase is intended to support continued development within the TETRA project toolbox.

Appendix A

Appendix

This appendix summarizes the main development decisions that led to the final framework. Where exploratory observations differ from the finalized evaluation, the results reported in Chapters 4–6 take precedence.

A.1 Development Progression

The project aimed to transform French hydro-ecological restoration reports, containing interleaved narrative text, maps, photographs, tables, and diagrams, into a multimodal knowledge graph. The central design question was whether image-derived information could contribute useful, visually grounded knowledge beyond a text-only representation. The final research questions, methods, results, and limitations are reported in the main chapters.

A.1.1 Initial Architecture and Extraction Revision

The initial prototype combined PyMuPDF-based extraction, CLIP-based image filtering, and zero-shot BLIP captioning. Early tests on real report PDFs revealed two main limitations: native PDF extraction fragmented complex maps and figures into multiple XObjects, and many extracted assets were non-informative logos, banners, or repeated graphics. BLIP-1 [20] and BLIP-2 [2] also produced generic captions and did not reliably interpret dense labels, tables, or technical diagrams.

The extraction stage was therefore redesigned around full-page rasterization, layout detection, structural filtering, and OCR-supported caption deduplication. This revision produced coherent figure-level assets suitable for downstream retrieval and captioning. The final extraction procedure and its evaluation are described in Sections 4.2 and 5.5.

A.1.2 Retrieval and Captioning Development

The initial CLIP matching approach was replaced by a comparative retrieval study covering document-specialised single-vector and multi-vector late-interaction models, including ColPali [6]. This transition reflected the need to associate figure-level visual content with relevant narrative blocks rather than rely on generic image–text similarity. Formal retrieval evaluation was subsequently introduced using Recall@5, Precision@1, MRR@5, and nDCG@5 [57]; the finalized ranking and retriever selection are reported in Section 5.6.

Captioning development similarly progressed from generic zero-shot models to instruction-following VLMs capable of interpreting technical visual content and retrieved French context. Early experimentation showed that placing retrieved text before visual observation could cause the model to confirm textual expectations rather than describe the image. This motivated the two-phase pixel-first prompt architecture: visual observation precedes contextual enrichment, and contextual statements are restricted to elements

already identified in the image. The final prompting, context-fusion, and hallucination-mitigation methodology is detailed in Section 4.4.

A.1.3 Context, Prompting, and Decoding Refinement

Several context and generation strategies were evaluated to balance visual grounding with information enrichment. Document-level summaries supplied broader thematic context but could omit image-specific detail; retrieved local blocks supplied detailed evidence but could be incomplete. Their blended fusion was therefore evaluated alongside local-only and parallel context configurations.

Prompt-level interventions subsequently tested stricter observation-first constraints, refusal of unsupported claims, negative-evidence checking, self-correction, and domain-persona framing. Contrastive decoding variants were also evaluated to suppress language-prior-driven generation by contrasting standard decoding with instruction- or image-perturbed decoding. The finalized prompt-engineering and contrastive-decoding procedures are described in Sections 4.4.6 and 4.4.7; their comparative results are reported in Sections 5.7.5 and 5.7.6.

A.1.4 Knowledge-Graph Construction and Fine-Tuning

The knowledge-graph stage evolved from direct Turtle generation to a two-step procedure in which the language model produces JSON triples and a deterministic RDFLib-based serializer validates and maps them to TetraOnto. This separation reduced serialization failures caused by malformed syntax and preserved provenance metadata for text- and image-derived information. Entity normalization and cross-modal resolution were added to support merging facts expressed differently across modalities.

The final stage evaluated LoRA-based adaptation of the triple-extraction model using a synthetic, schema-filtered French instruction corpus. Fine-tuning improved predicate-vocabulary conformance and altered cross-modal graph structure, while producing fewer triples than the baseline. The resulting structural trade-offs and the absence of a completed factual triple-precision audit are reported in Section 5.8.2.2.

A.2 Summary

The final system emerged through successive revisions to extraction, retrieval, caption generation, and RDF construction. Each major change addressed a specific limitation observed during development: fragmented visual assets, structural document noise, weak generic captioning, context-induced hallucination, or fragile RDF serialization. The final four-pipeline architecture and its limitations are evaluated in Chapters 4–6.

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